Fundamentals of Robotics and Applications

Unit 2: Robot Anatomy and Motion Analysis

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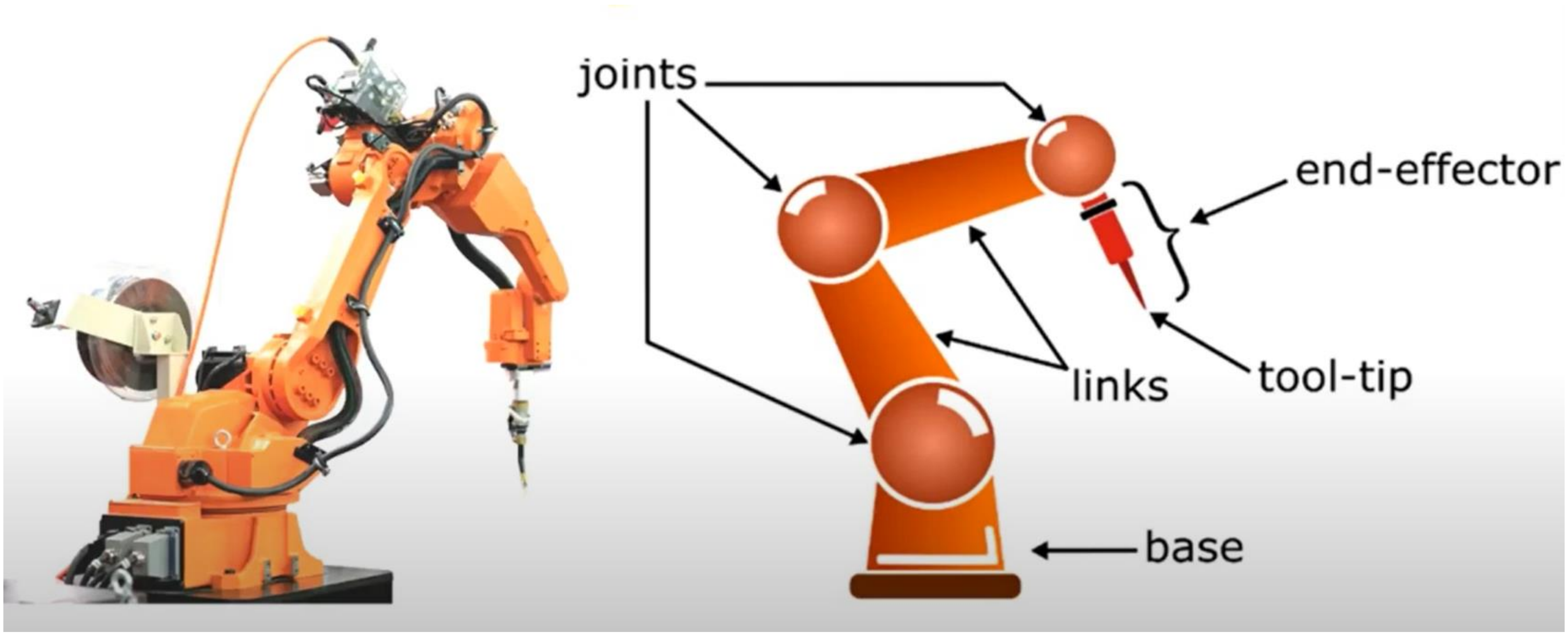
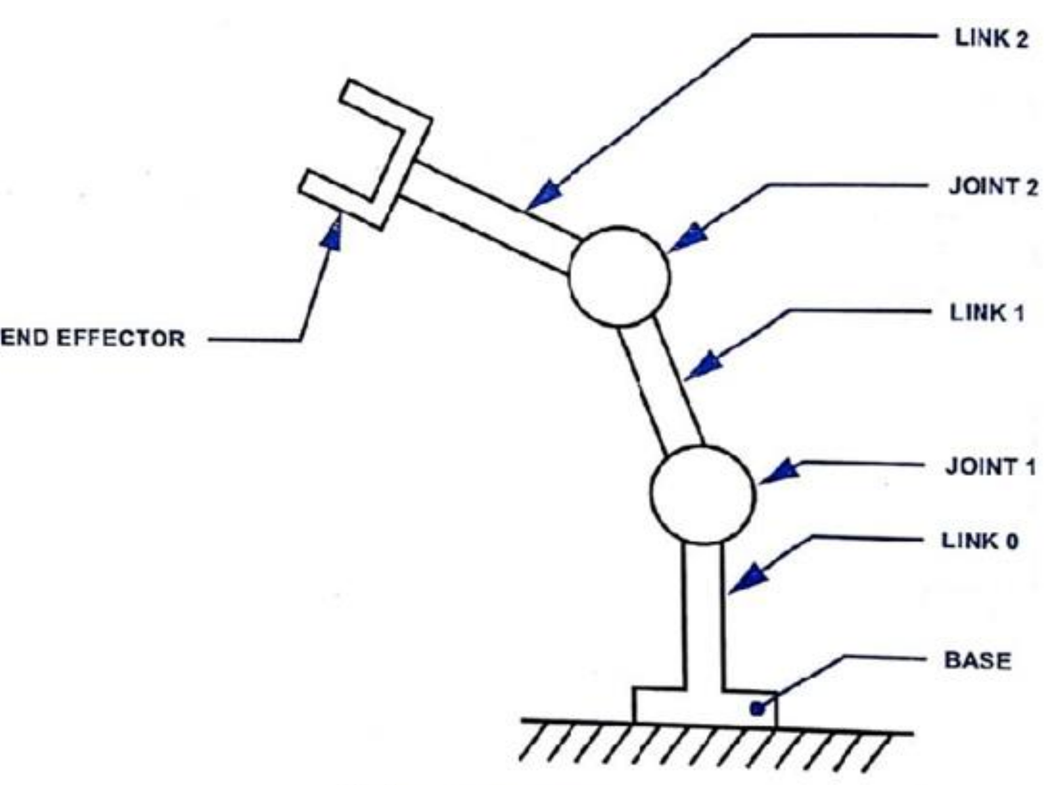
Robot Anatomy



Robot Anatomy



Simple Robot Construction

Robot anatomy is concerned with the physical construction of the body, arm and wrist of the machine.

The body is attached to the base and the arm assembly is attached to the body.

At the end of arm is the wrist.

Wrist consists of number of components that allow it to be oriented in a variety of positions.

Relative movements between various components of body, arm and wrist are provided by series of joints.

The body, arm, and wrist assembly is called as manipulator.

Attached to the robots wrist is a hand or a tool called the end effector. End effector is not considered as part of robot anatomy.

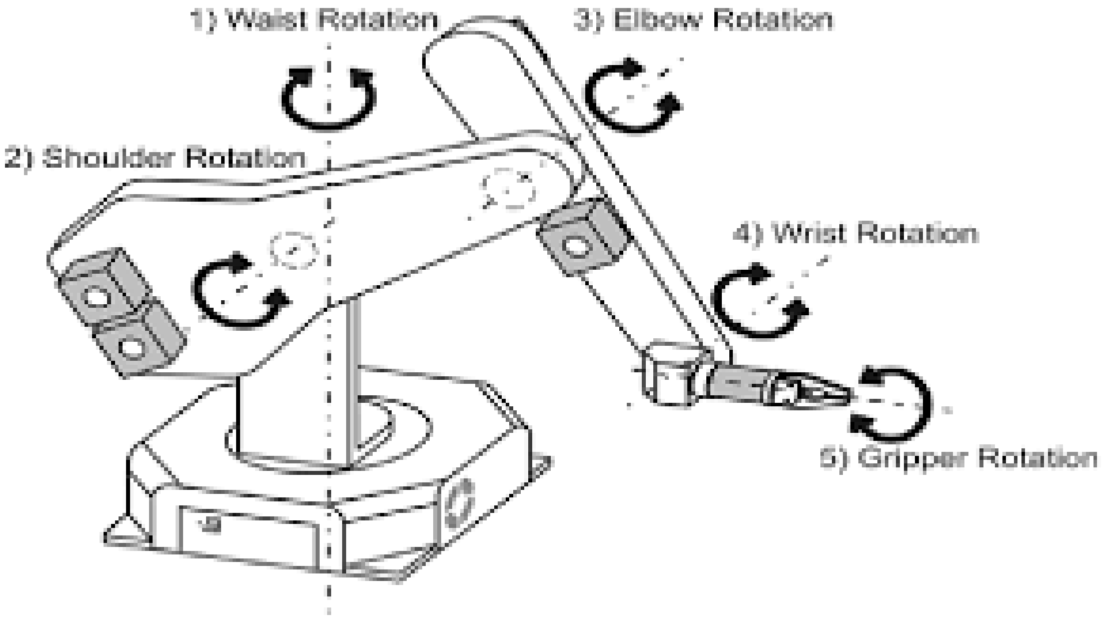
The arm and body joints of the manipulator are used to position the end effector and the wrist joints of the manipulator are used to orient the end effector

Robot Anatomy

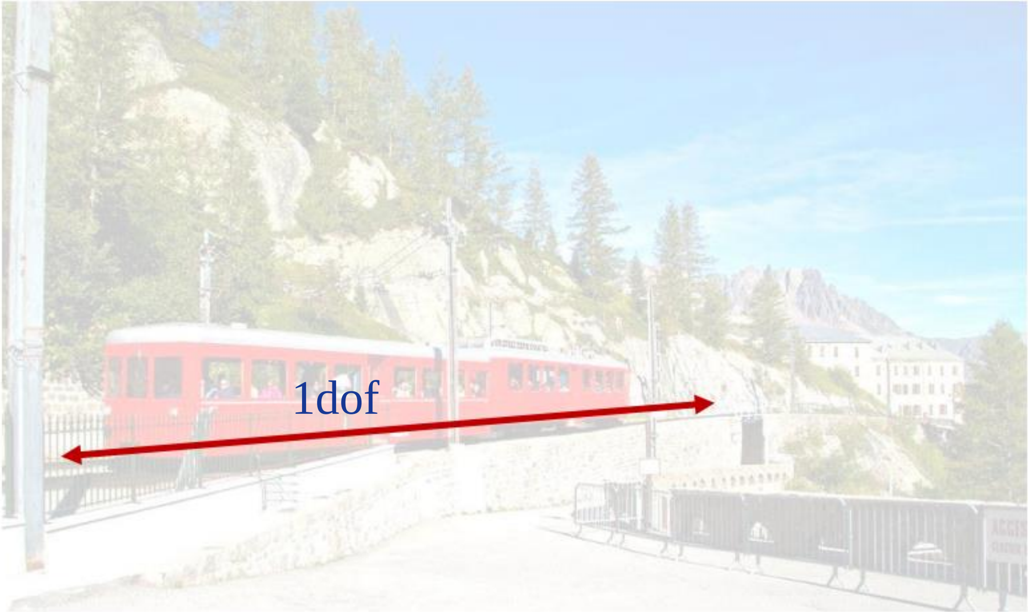
- **Joint:** A joint is the one that integrates two or more links to provide controlled relative movement between input link and the output link.
- **Link:** The link is a rigid member that connects the joints. Link can be an input link or an output link. The movement of the input link causes various motions of the output link.
- **End effectors:** End effectors or end-of-arm tool is the device at the end of the robotic arm which is shaped like a hand or as a special tool depending upon the application.
- **Base**: The support for the robot arm is called as the base.

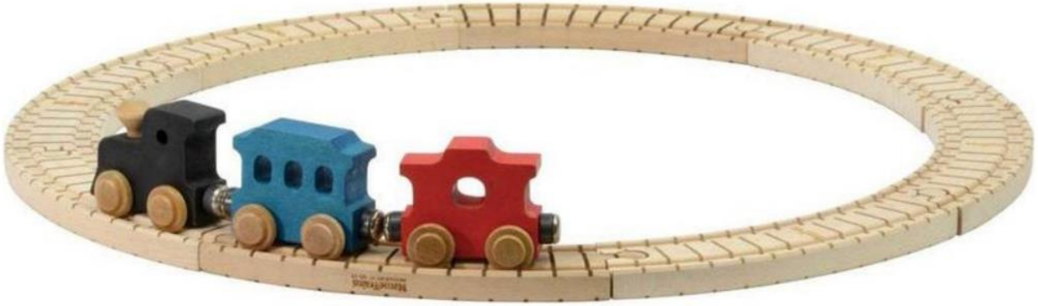
Degrees of freedom

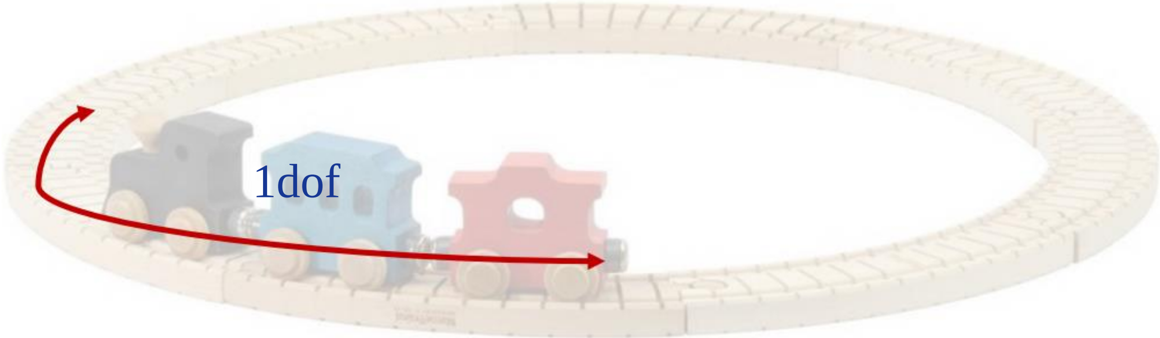
- **Degrees of freedom (d.o.f):** The degrees of freedom describe a robot's freedom of motion in the three-dimensional space.
- The degrees of freedom of a robot typically refer to the number of movable joints of a robot.
- A robot with three movable joints will have three axis and three degrees of freedom, a four axis robot will have four movable joints and four axis, and so on.



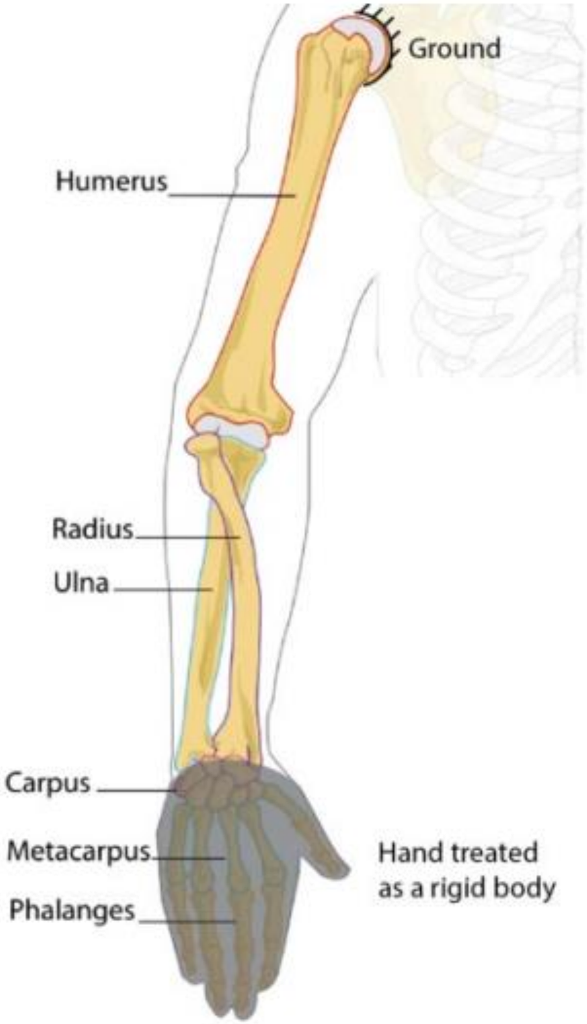
How many degrees of freedom does this have?



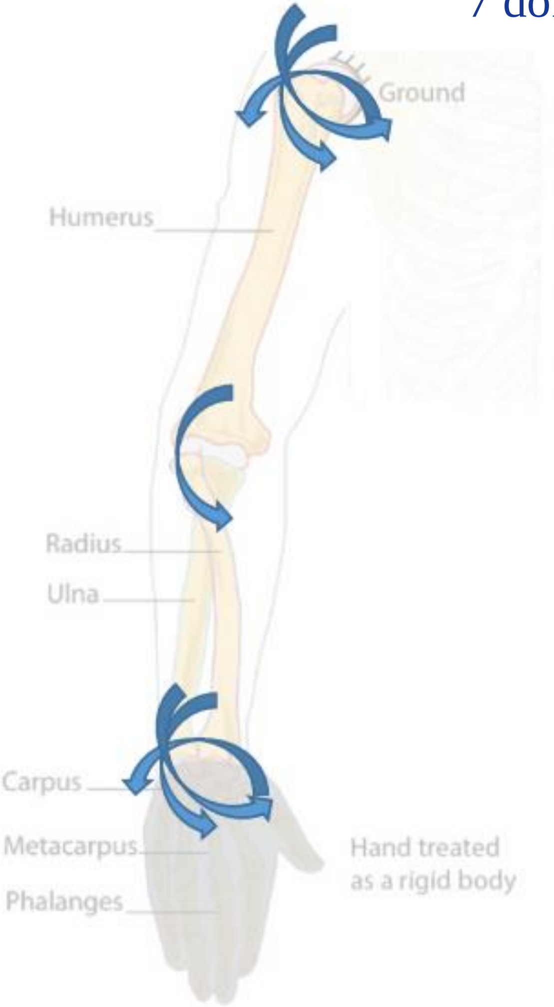




How many degrees of freedom does this have?

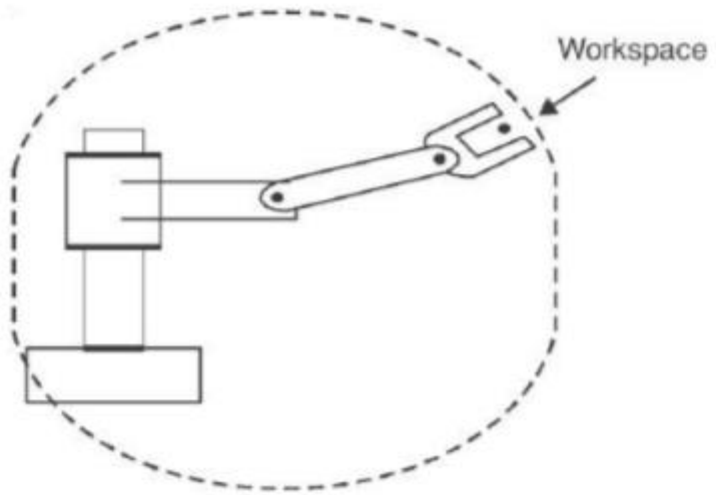


7 dof



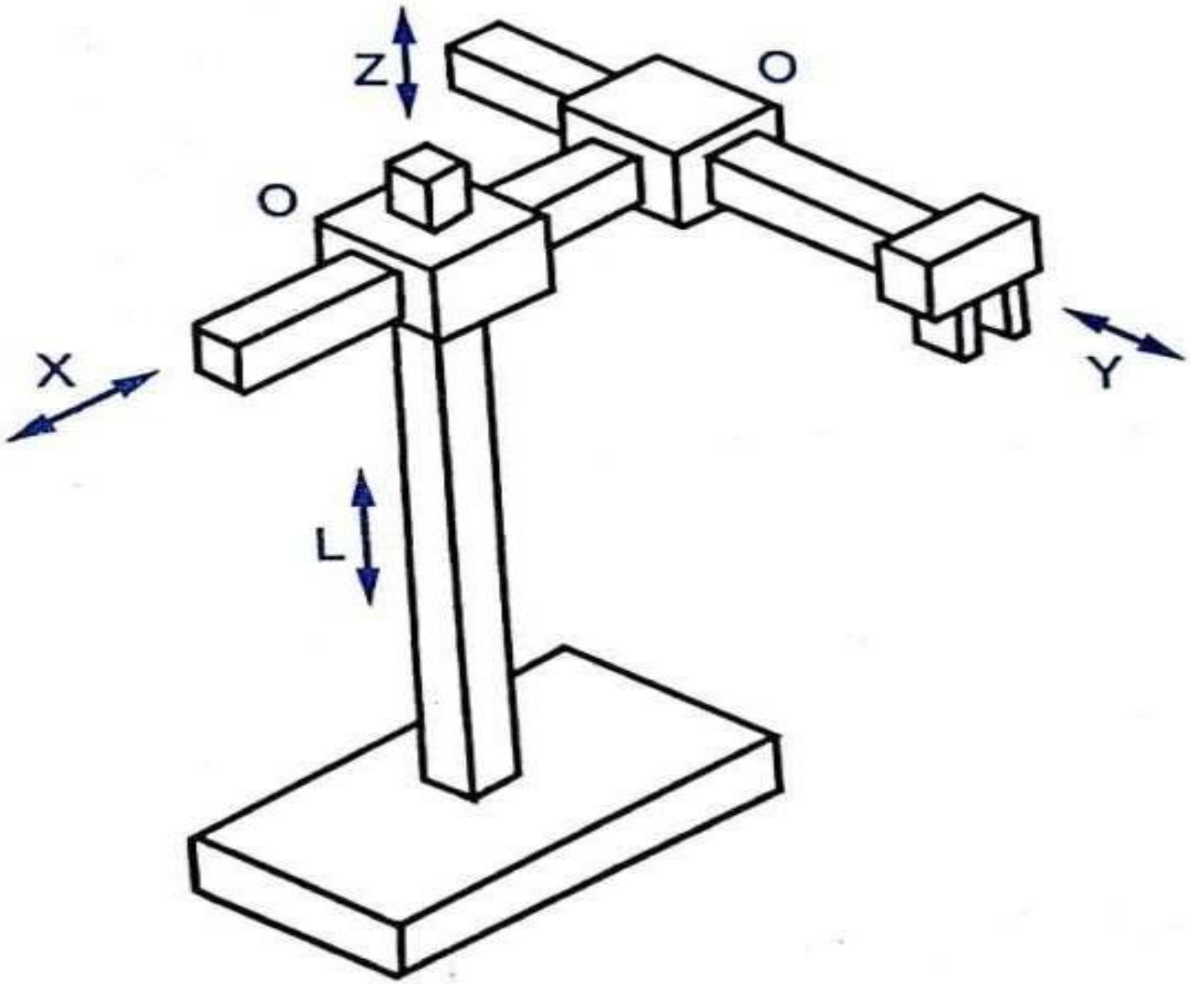
Wok volume/envelope

- **Work envelope or work volume** A robot can only work in the area in which it can move. This area is called the work envelope.
- The work envelope is determined by how far the robot's arm can reach and how flexible the robot is.
- The more reach and flexibility a robot has, the larger the work envelope will be. It is one of the most important characteristics to be considered in selecting a suitable robot



Robot configurations

- **Cartesian Co-ordinate Robot**



Cartesian Co-ordinate Robot

It is also called as a rectilinear robot or a XYZ Robot. It consists of Three prismatic joints along the X, Y and Z direction in three dimensional spaces. The motion of tool tip is smoother.

Advantages:

Allows for simple controls.

Posses high degree of mechanical rigidity, accuracy repeatability.

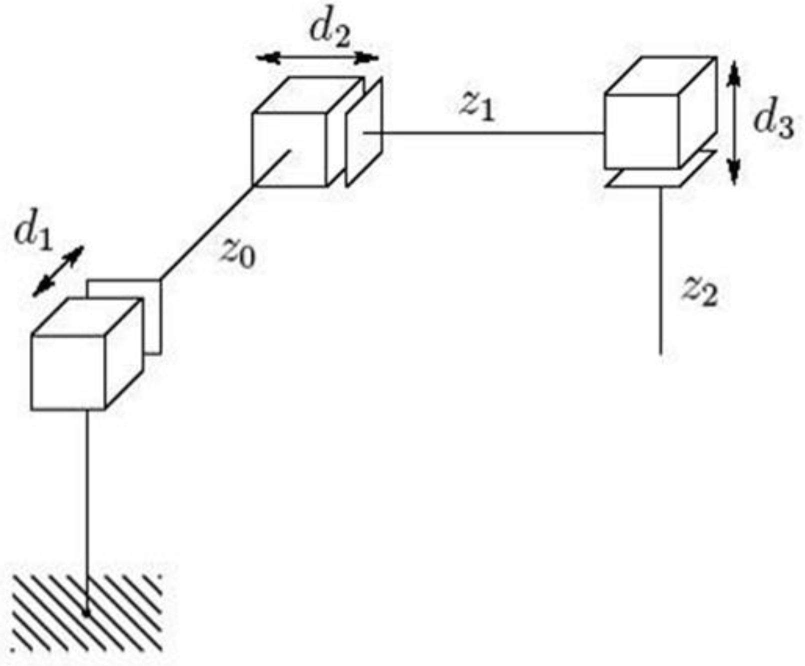
Disadvantages:

Limited in their movements to a small rectangular work space.

Reduced Flexibility.

Applications:

Pick and Place applications, Material handling, loading and unloading of machining operations.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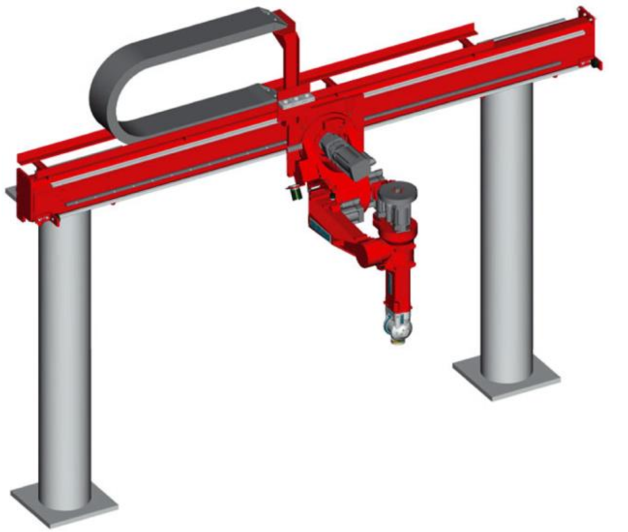
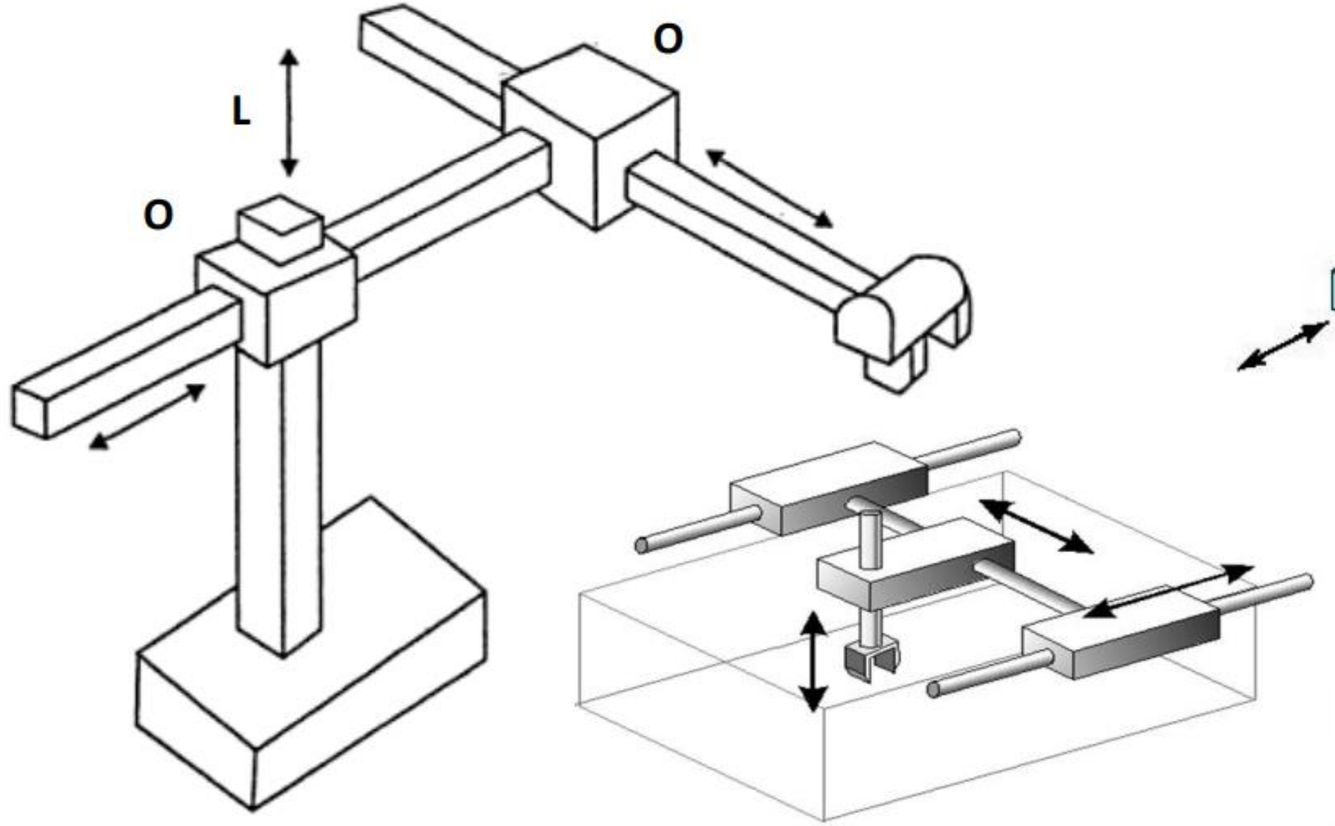
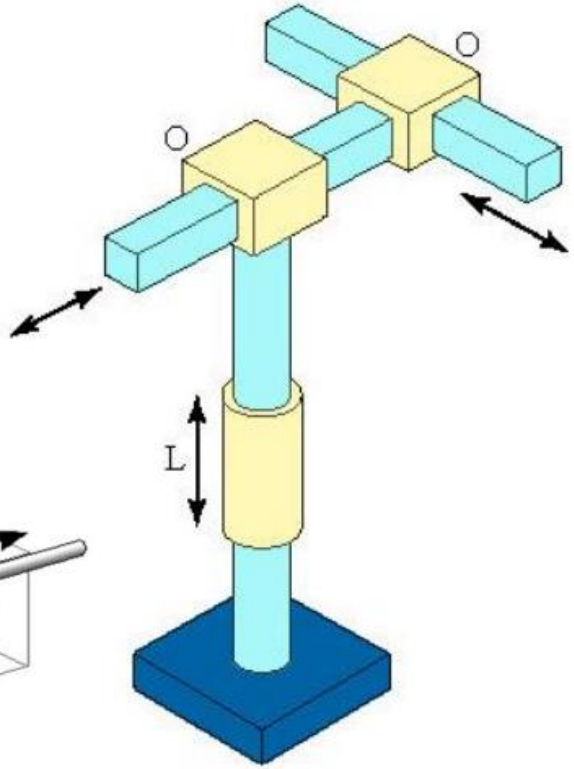
EPSON

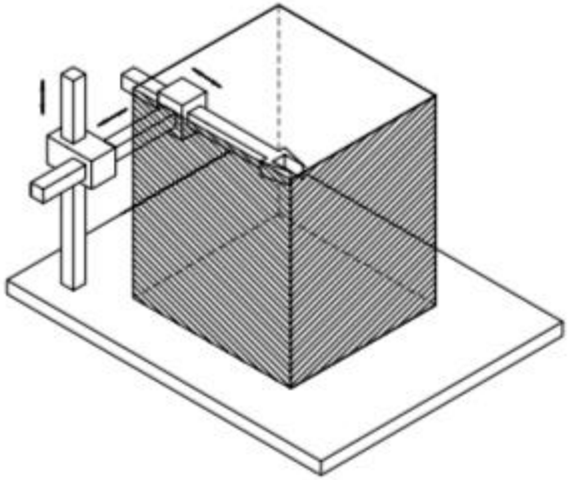


Wok volume/envelope

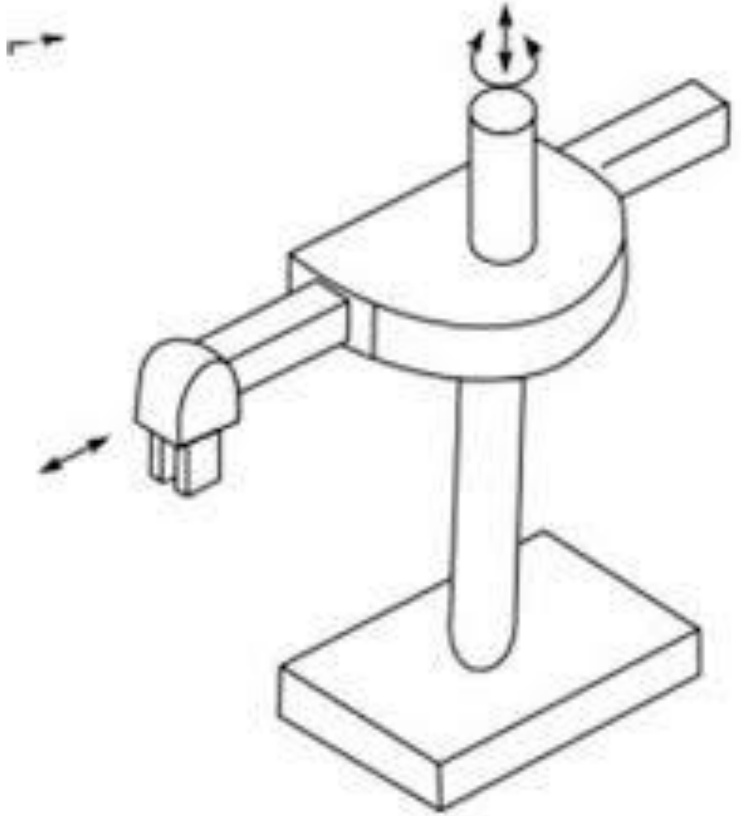
- **Cartesian Configuration Robots:** Configuration provides rectangular work envelope







- **Cylindrical Configuration**



Robots of the cylindrical configuration consists of a slide in the horizontal position and a column in the vertical position.(**consists of 2 linear movement and 1 rotational)**

Advantages:

Rigidity is increased and is quite Robust

Has capacity to carry heavy payloads.

Disadvantages:

Work volume is less

Occupies more floor space.

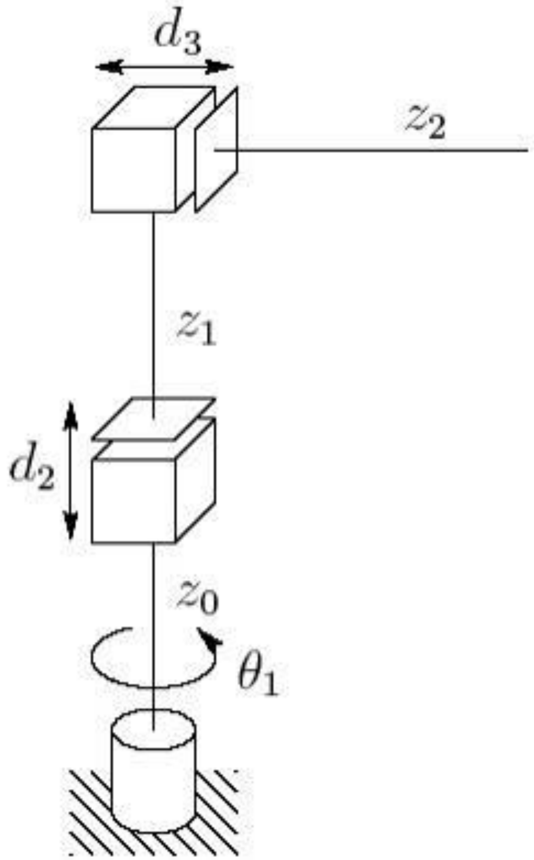
Applications:

Foundry, forging applications.

Casting, Conveyor pallet transfers.

Machine Loading and unloading.

- X = horizontal rotation of 360°, left and right motions
- Y = vertical, up and down motions
- Z = horizontal, forward and backward motions


d_3
z_2
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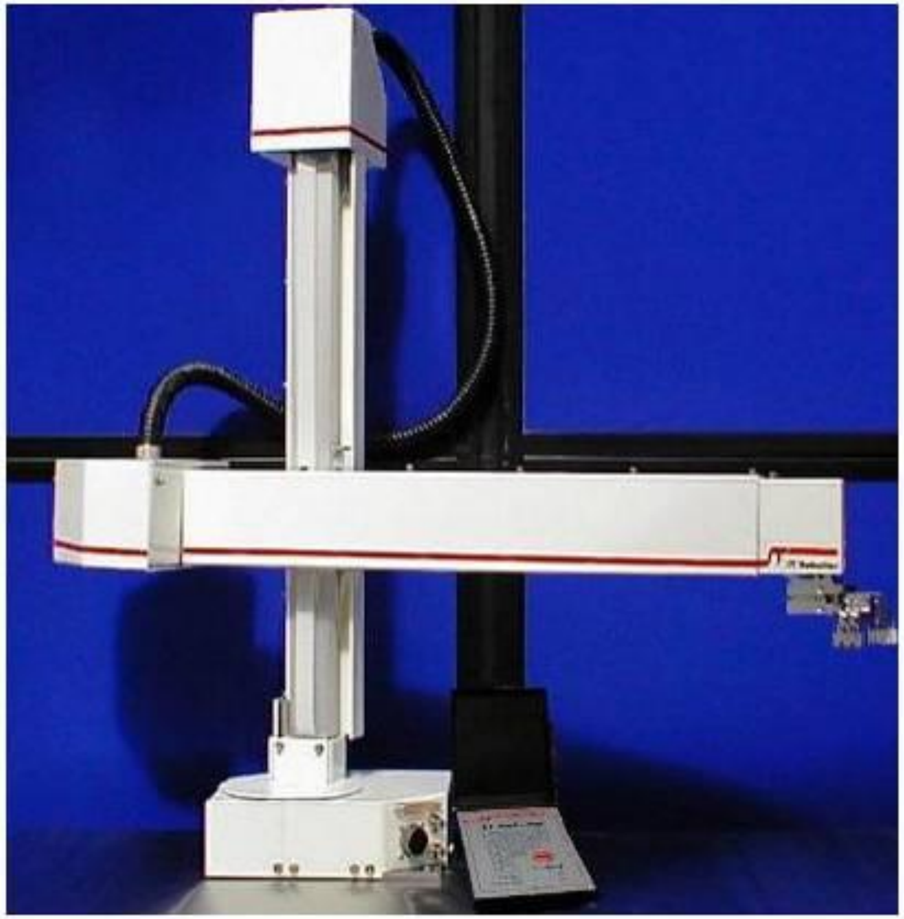
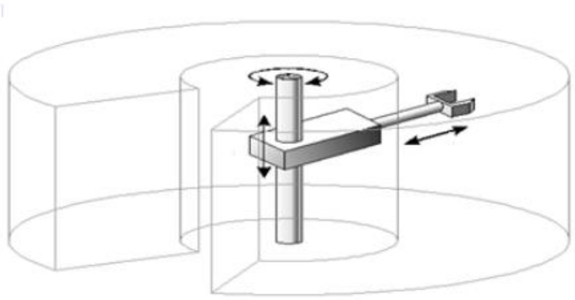
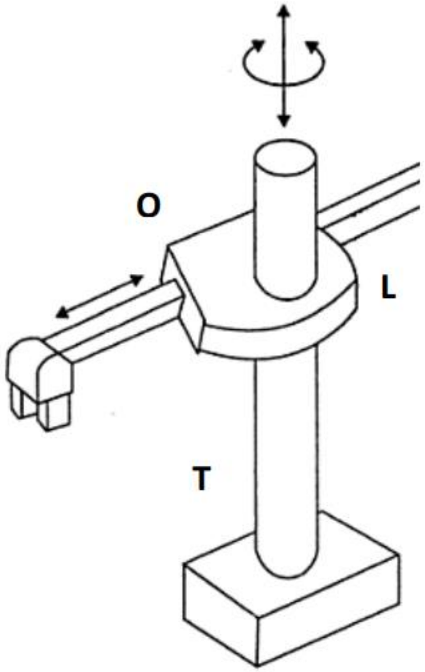


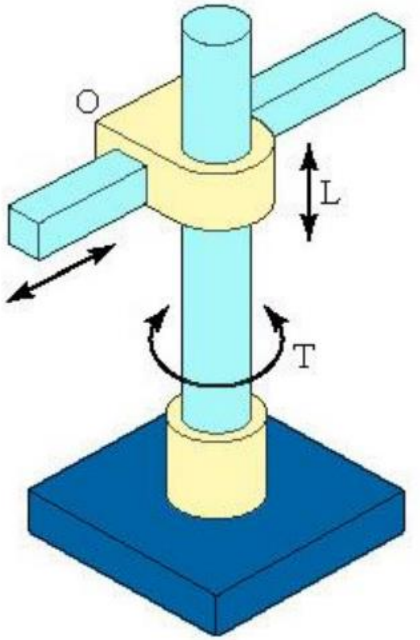

GMFRobotics
M-100

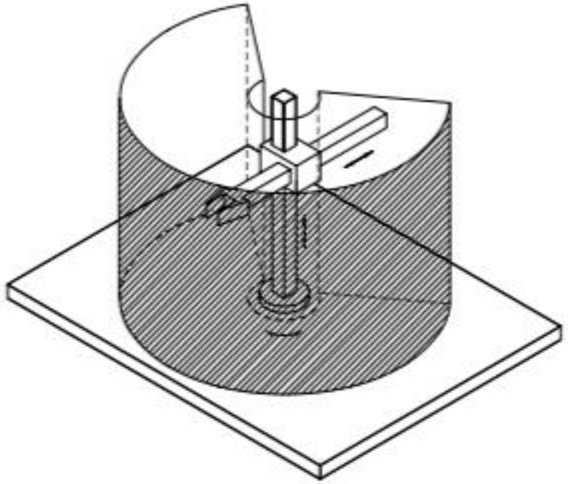
Wok volume/envelope

- **Cylindrical Configuration Robots:** Configuration provides cylindrical work envelope, has good work area to floor area ratio

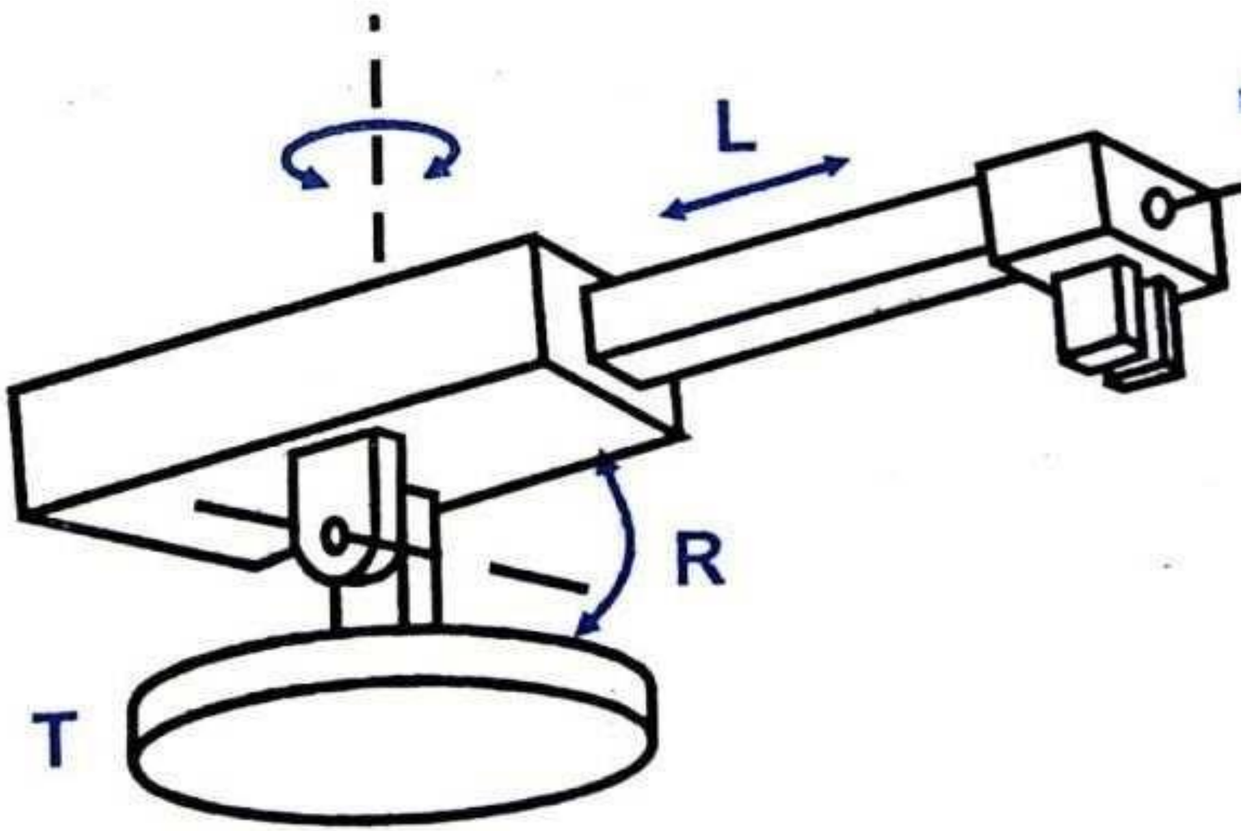








- **Polar Configuration (Spherical Configuration)**



Polar Configuration Robot

It consists of 1 linear movement (L) and 2 rotational joints(R Joint-rotate about horizontal axis) and (rotate about Vertical axis T).

Advantages:

Long reach capabilities.

Disadvantages:

The vertical reach is low

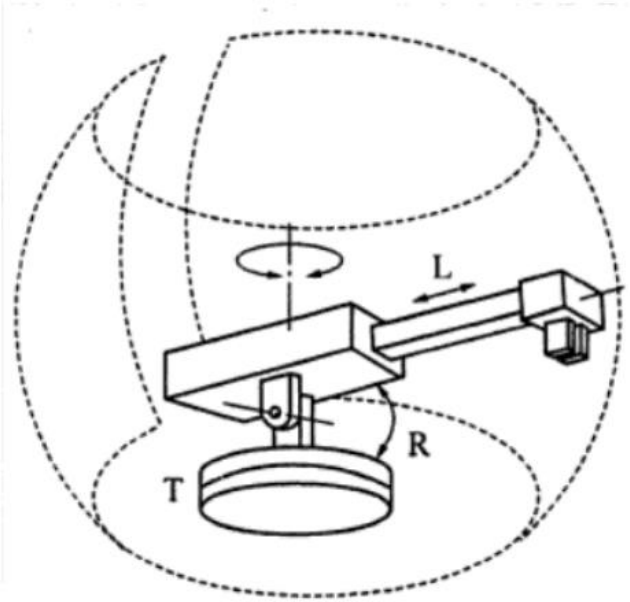
Applications:

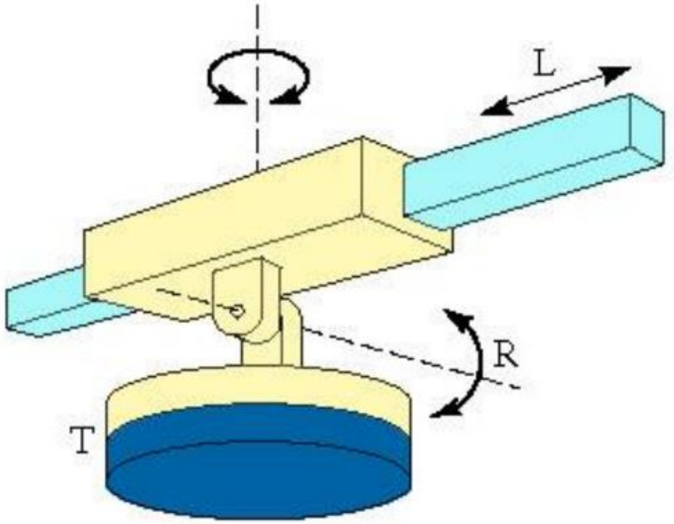
Die casting, forging, injection moulding, Dip coating, cleaning of parts.

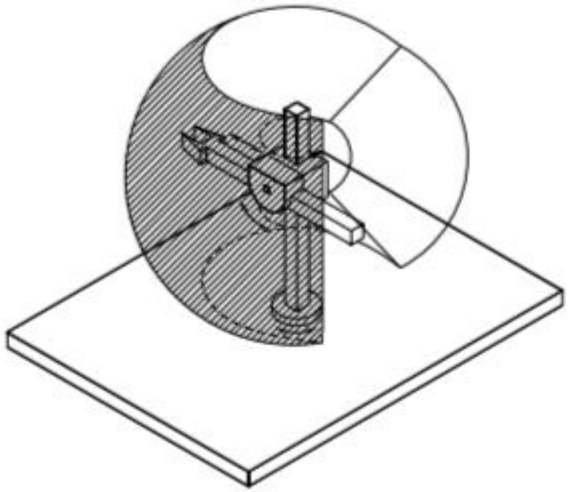
- X = horizontal rotation of 360°, left and right motions
- Y = vertical rotation of 270°, up and down motions
- Z = horizontal, forward and backward motions

Wok volume/envelope

- **Polar (Spherical) Configuration Robots:** Configuration provides spherical work envelope

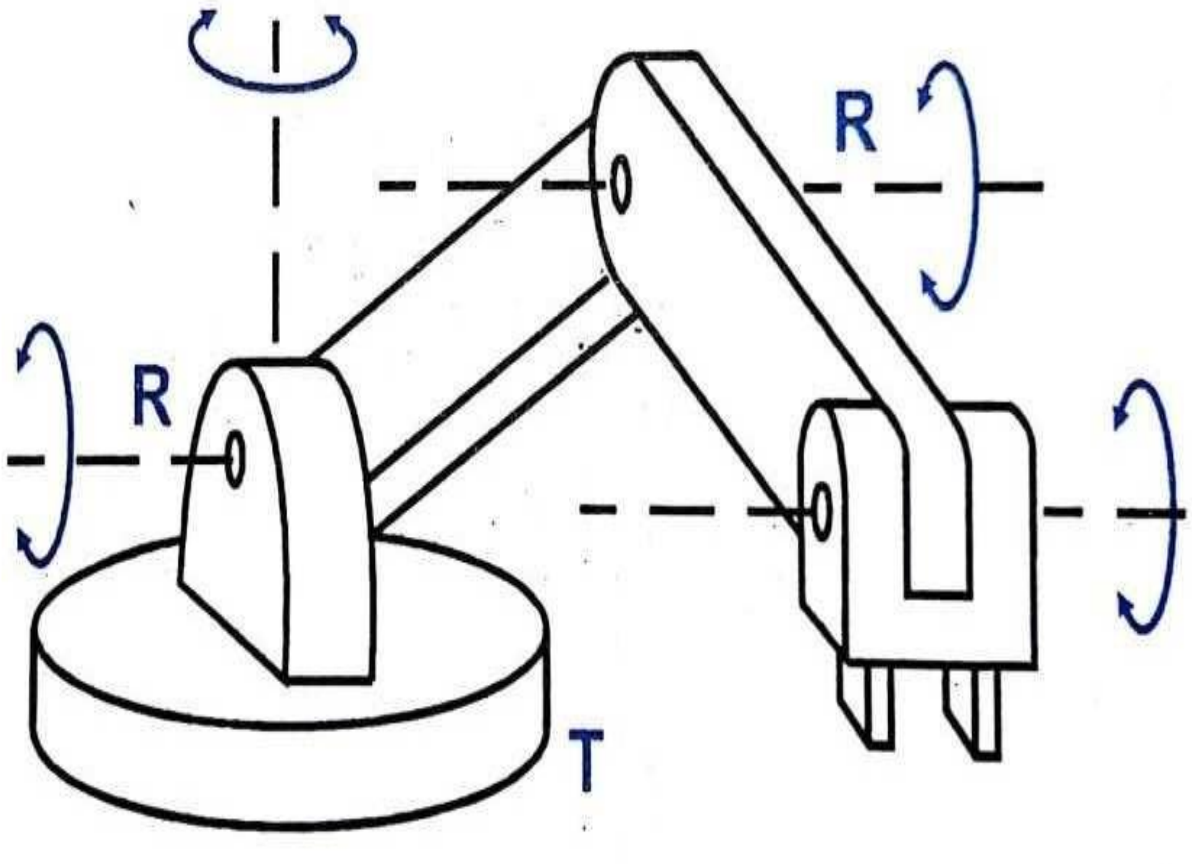






Robot configurations

- **Jointed-arm Configuration Robot**



Jointed arm Configuration Robot

This type of Configuration resembles the human arm where the column swivels about a base.

These robots have 4 to 6 axes.

All joints are rotary joints and wrist will be attached at end.

Advantages:

Work volume available is large.

Operations is quick.

Flexibility is increased.

Disadvantages:

Operating procedures are difficult

Quite expensive type of configurations.

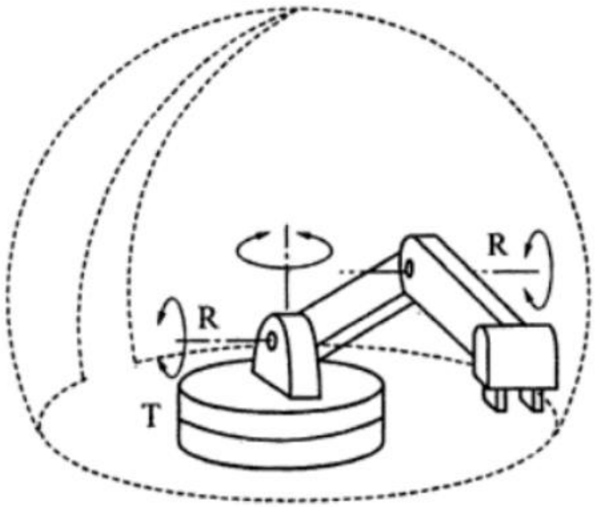
Number of components involved are more.

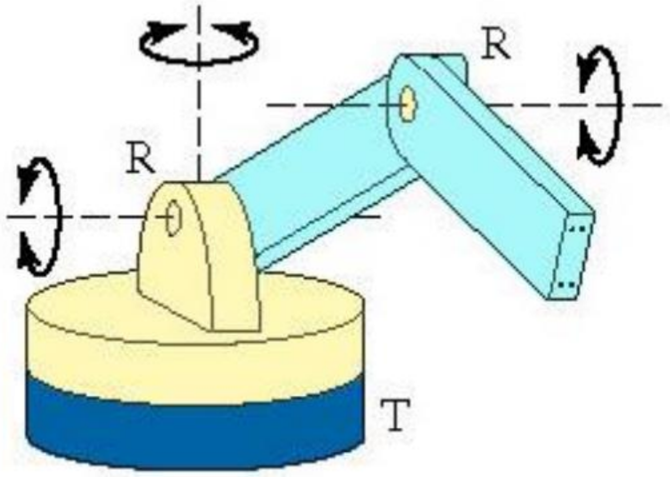
Applications:

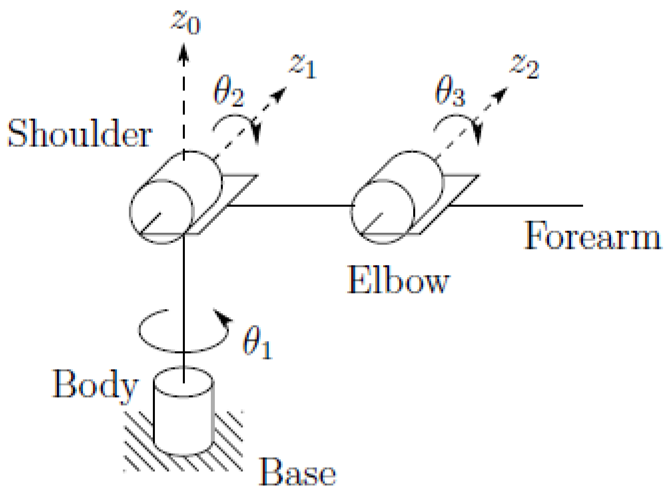
To perform welding, spot welding, spray painting operations.

Wok volume/envelope

- **Revolute Robots:** Configuration is similar to that of human arm, Consists of two straight links, corresponding to the human forearm and upper arm, connected by a rotary joint.
- Provides spherical work envelope, has excellent work areas to floor area ratio



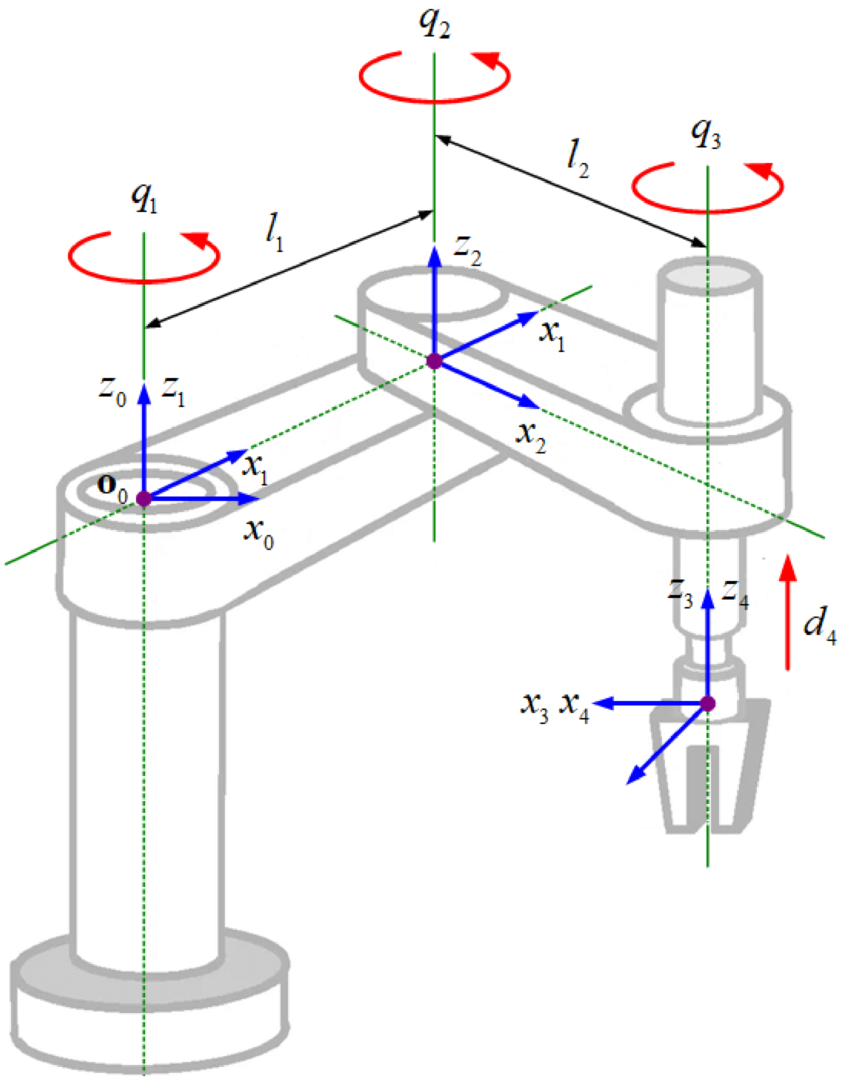



z_0
θ_2
z_1
θ_3
z_2
Shoulder
Forearm
Elbow
θ_1
Body
Base


UNIMATE
PUMA 500


KUKA
KUKA

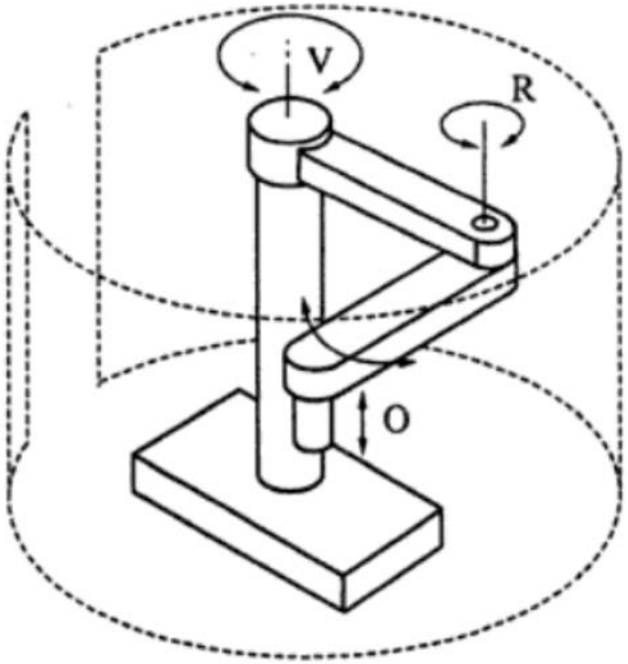
SCARA(Selective Compliance Assembly Robot Arm /Selective Compliance Articulated Robot Arm)

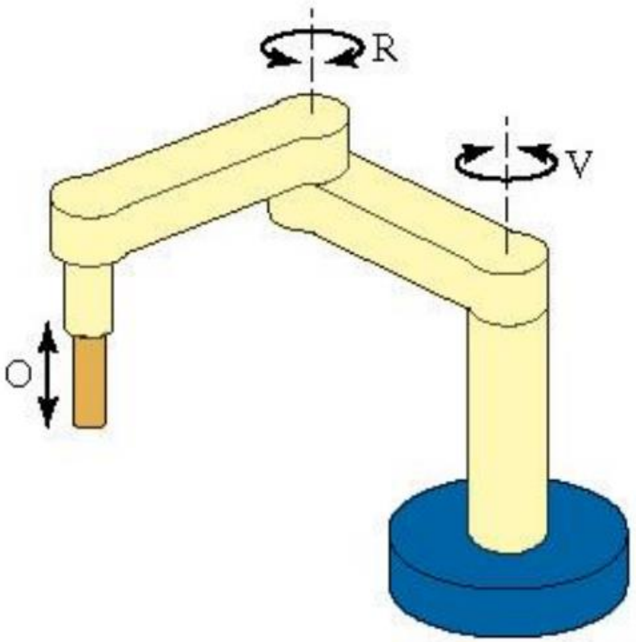


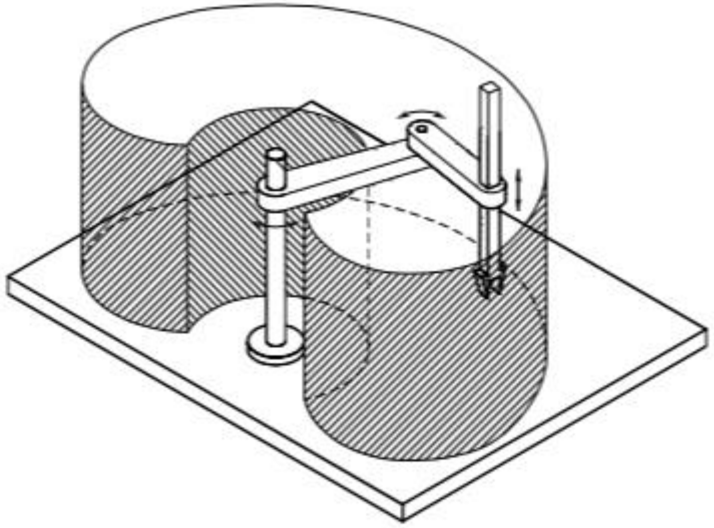
- Provides one linear and three rotary motions (4dof).
- Provides cylindrical work envelope with high speed drive motors.
- Unique configuration and designed to handle a variety of material handling operations.
- These are popular option for small robotic assembly applications.
- Commonly used for pick and place or assembly operations where high speed and high accuracy are required.

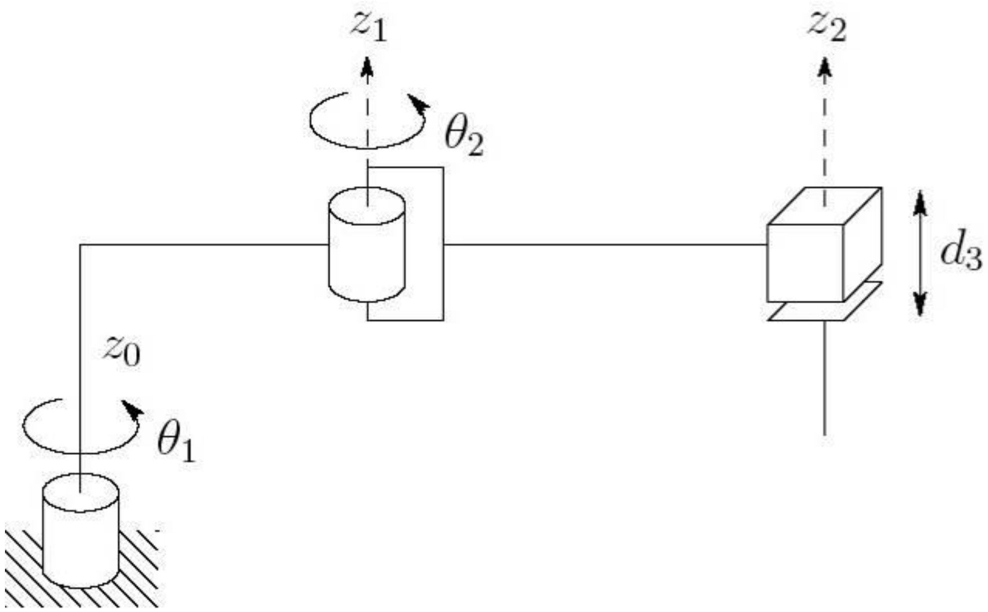
Wok volume/envelope

- **SCARA (Selective Compliance Assembly Arm) Robots:** Usually, this robot has 2 parallel rotary joints to provide compliance in a plane. Provides cylindrical work envelope with high speed drive motors.



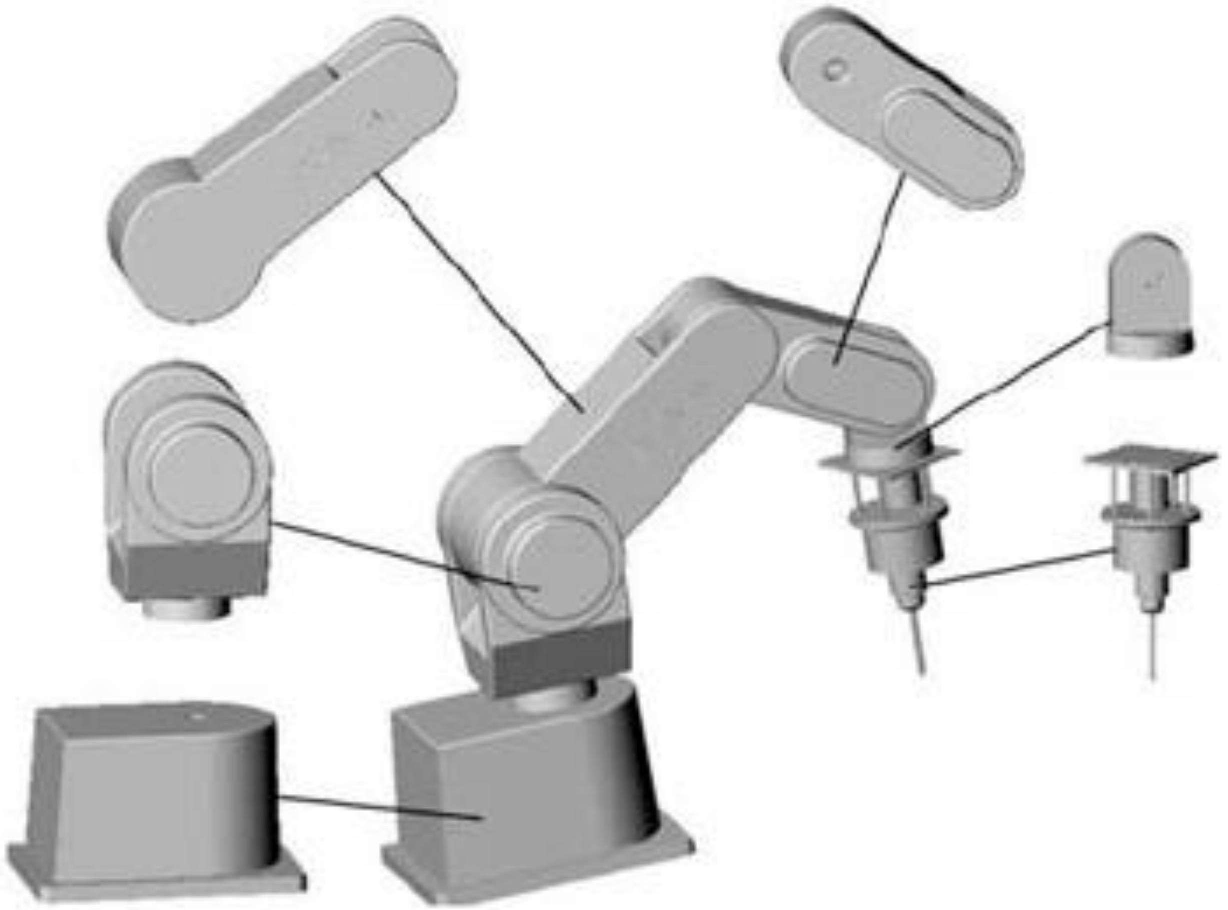






https://www.youtube.com/watch?v=_canCYWZPsc

Robot joints

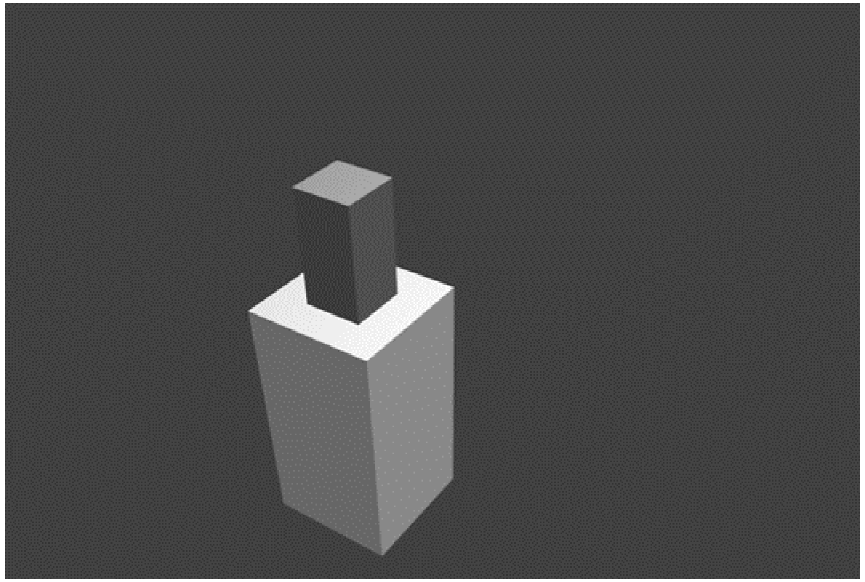


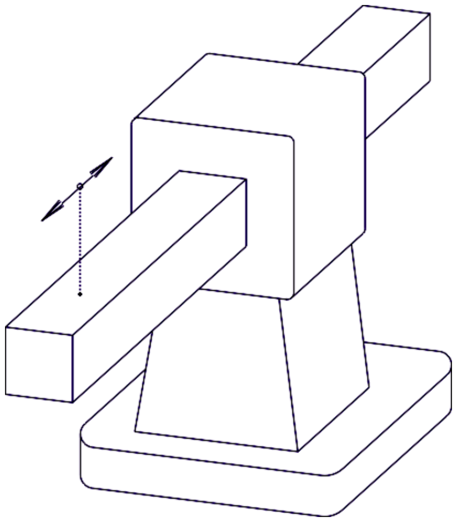
- Robot joints enable movement in robots by connecting two rigid links.
- The type of joint determines the range and nature of movement possible.
- In this context, links are used to refer to the rigid members connecting the joints for clear and adequate functioning.

Types of joints

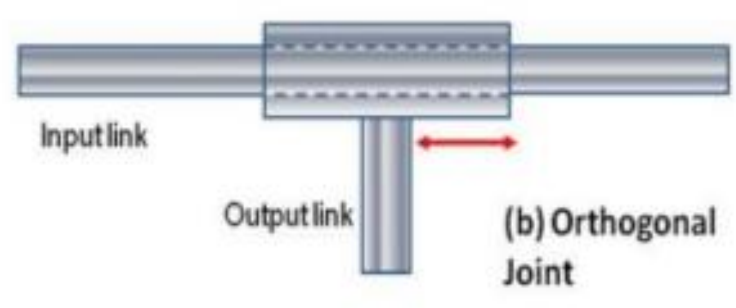
Linear joint or Prismatic joint (type P joint) The relative movement between the input link and the output link is a translational sliding motion, with the axes of the two links being parallel.

(a) Linear Joint

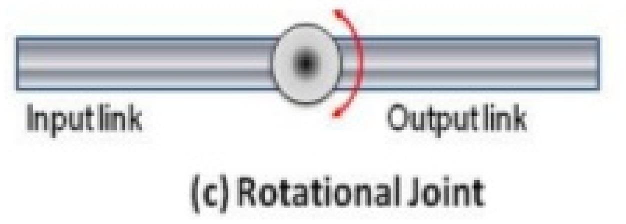




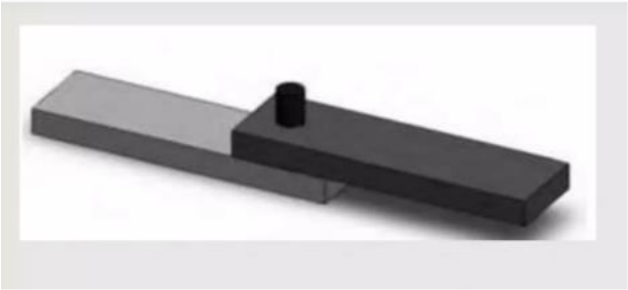
Types of joints

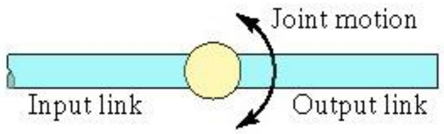


Orthogonal joint (type U joint) This is also a translational sliding motion, but the input and output links are perpendicular to each other during the movement.

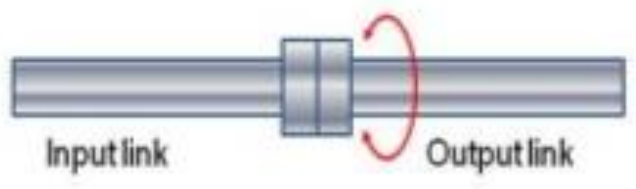


Rotational joint (type R joint) This type provides rotational relative motion, with the axis of rotation perpendicular to the axes of the input and output links.

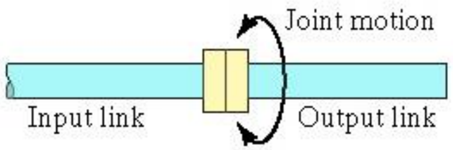




Types of joints

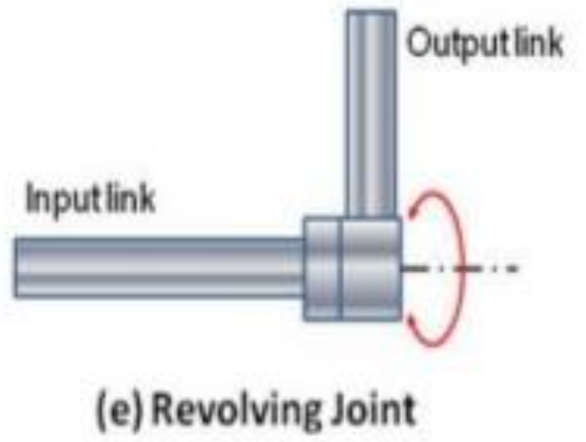


(d) Twisting Joint

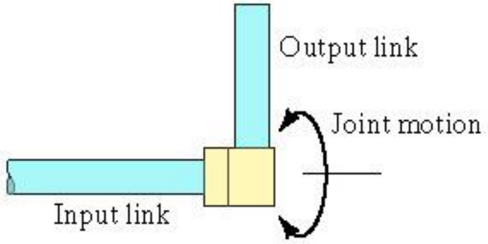


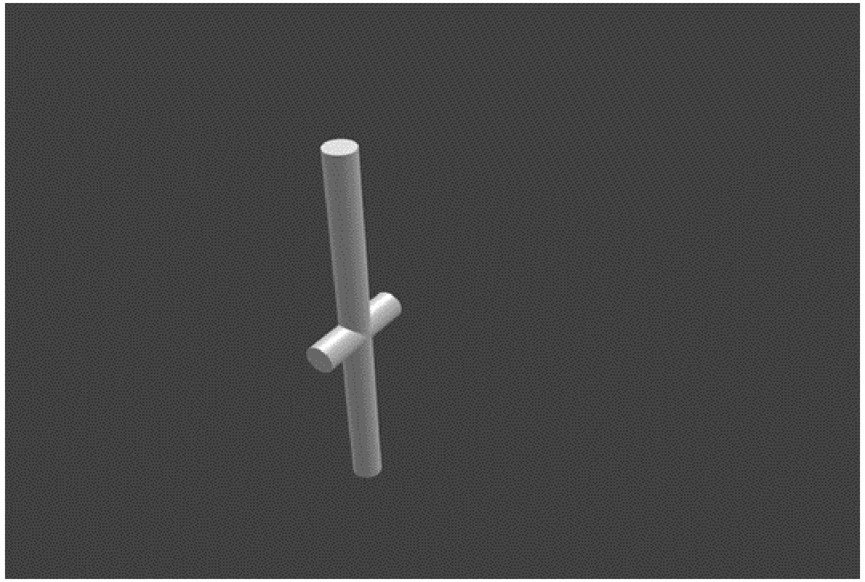
Twisting joint (type T joint) This joint also involves rotary motion, but the axis or rotation is parallel to the axes of the two links.

Types of joints

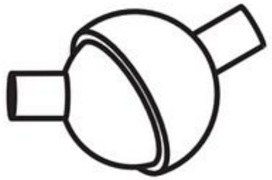


Revolving joint (type V-joint, V from the "v" in revolving) In this type, axis of input link is parallel to the axis of rotation of the joint. However the axis of the output link is perpendicular to the axis of rotation.



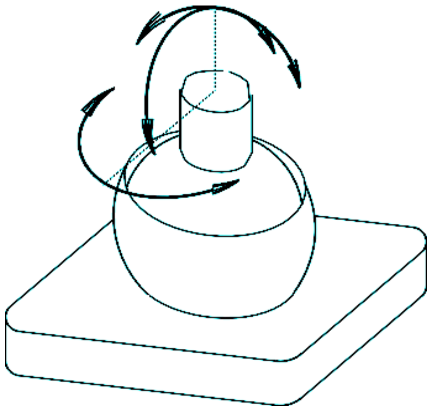


Types of joints



Spherical
(S)

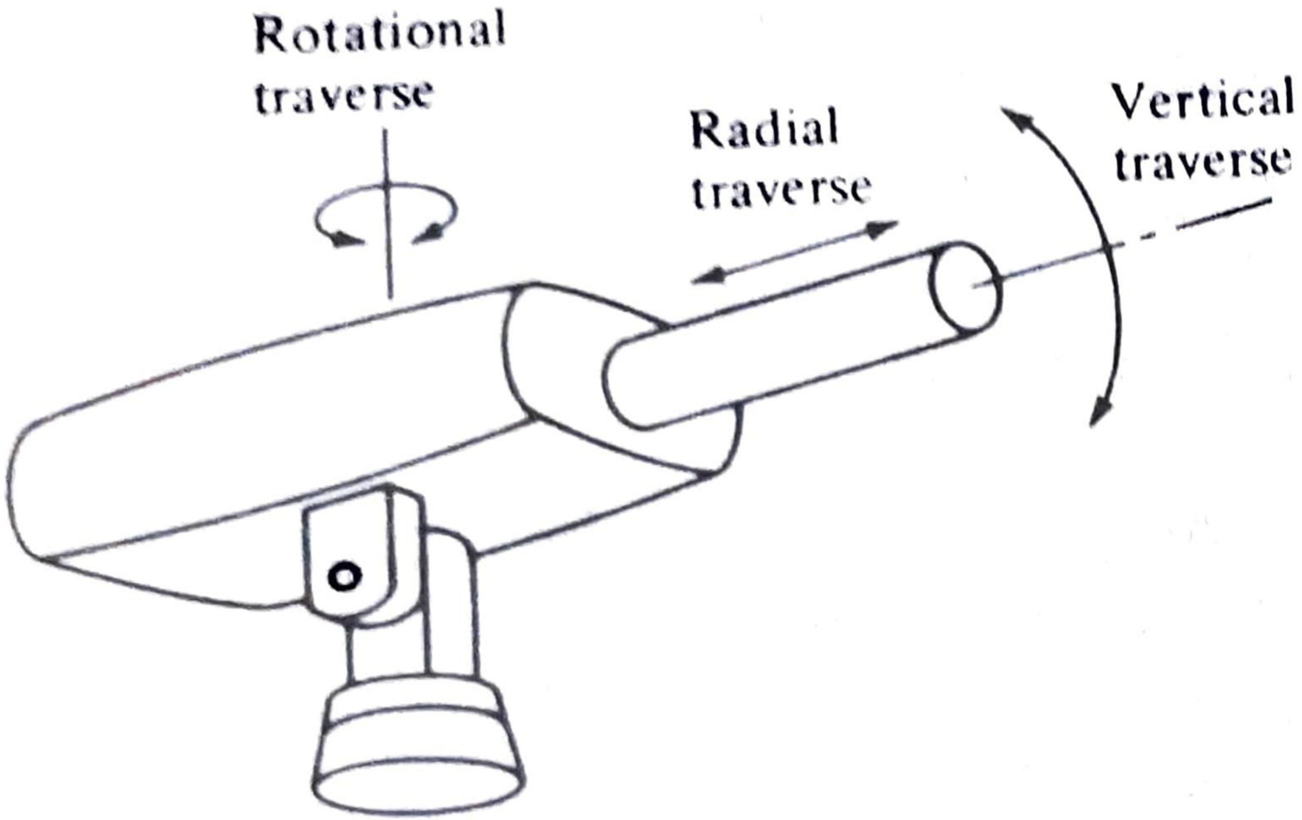
Spherical joints, also known as ball-and-socket joints, are a key component of robotic manipulators, often used in the shoulder and wrist. They allow for three degrees of rotational freedom around a central point, allowing robots to move in multiple directions at once, such as rotating, pivoting, and swiveling.



Types of movements (Robot Motions)

Arm motion & Body motions

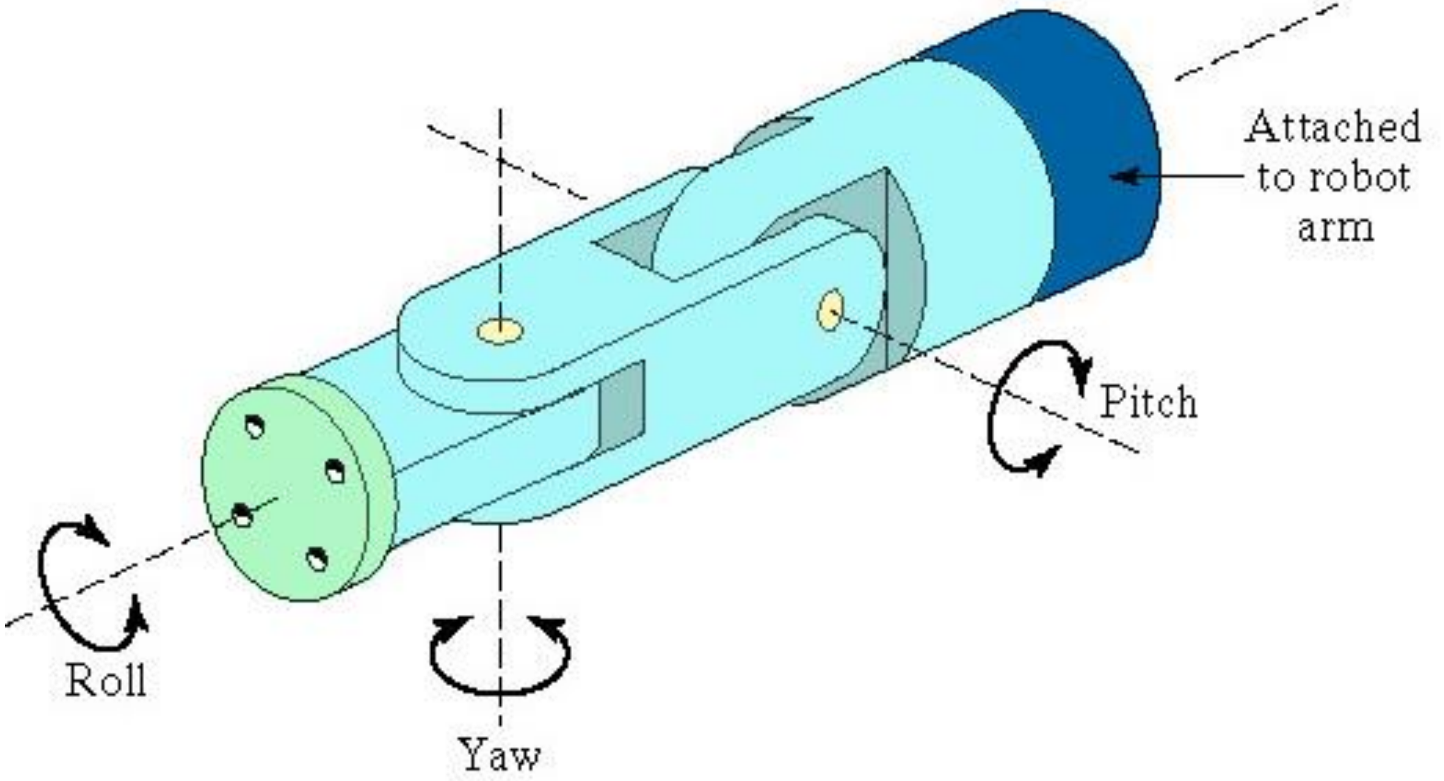
- **Vertical traverse:** This is the capability of the robot to move the wrist up and down to provide desired vertical altitude.
- **Radial traverse:** This involves extension and retraction (in or out movement of the arm) from the vertical center of the axis.
- **Rotational traverse:** This is rotation of the arm about the vertical axis.



Types of movements (Robot Motions)

Wrist motions

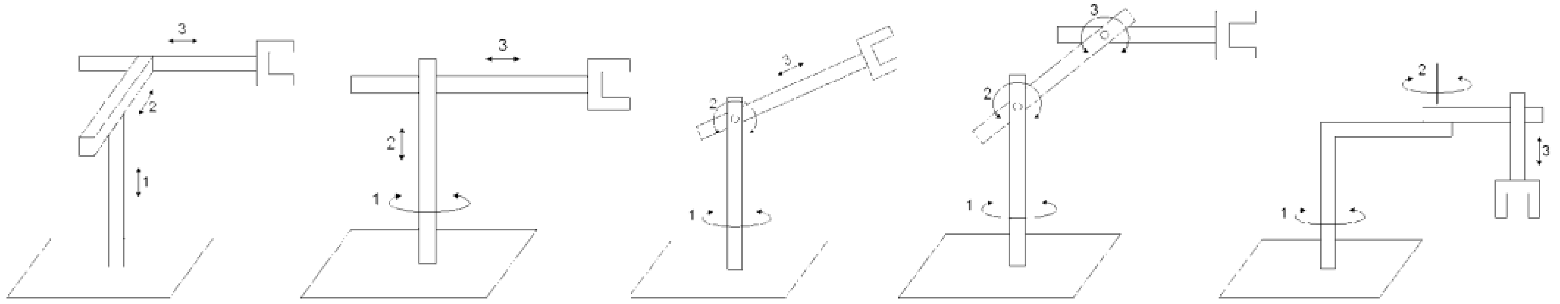
- **Wrist roll:** Also called as wrist swivel, the wrist roll involves rotation of the wrist mechanism about the arm axis.
- **Wrist Pitch:** Given that the wrist roll is in its center position, the pitch involves up and down rotation of the wrist, it is also referred as wrist Bend.
- **Wrist Yaw:** It involves the right and left rotation of the wrist



Basic Configurations of ROBOT

Configurations	Manipulator	Motions	Work Envelope	Application
Cartesian	LOO	3 Linear	Rectangular	Assembly, Palletizing Machine Tool Loading
Cylindrical	TLO	2 Linear , 1 Rotary	Cylindrical	Loading & Unloading on Machine Tools
Spherical (Polar)	TRL	1 Linear , 2 Rotary	Spherical	Spot Welding , Handling of Heavy Loads.
Revolute	TRR	3 Rotary	Spherical	Spray Painting, Seam Welding, Spot Welding, Assembly, Heavy Material Handling
SCARA	VRO	1 Linear, 2 Rotary	Cylindrical	Assembly Operations

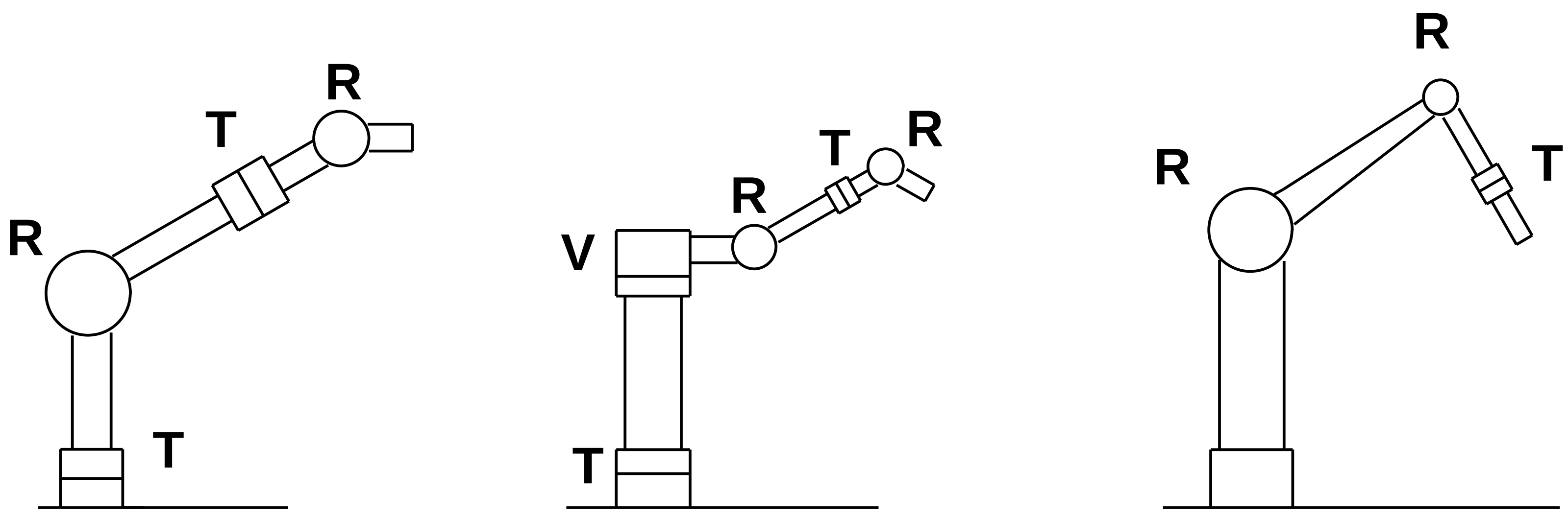
L = Linear; **O** = Orthogonal; **R** = Rotational; **T** = Twisting; **V** = Revolving



Example

- Sketch following manipulator configurations
- (a) TRT:R, (b) TVR:TR, (c) RR:T.

Solution:



(a) TRT:R (b) TVR:TR (c) RR:T

Design considerations of Links & Joints

- Strength and stiffness
- Material used for fabrication
- Weight and inertia
- Location of bearings/ selection
- The fits and tolerances of joints
- The external shape and aesthetics
- The friction and lubrication

How to decide

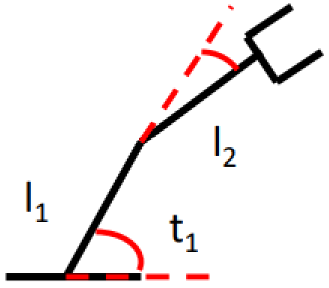
- What should be the link lengths of a robot?(L1=L2)
- Link velocities during control ??
- Position and Orientation of end effector?

Manipulation and Kinematics

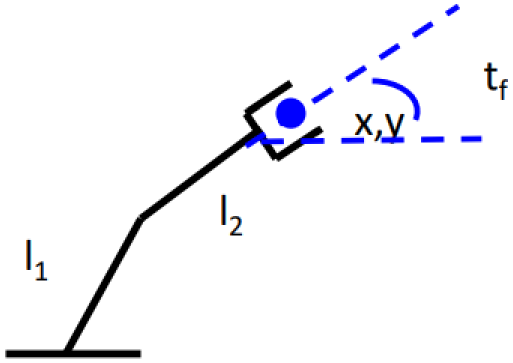
- The robot anatomy starts with the explanation of the mechanical manipulator, the motion of the manipulator and the understanding of the robot arm kinematics.
- Manipulation is the study of the relative motion giving the relationship between the objects themselves or the objects and the manipulator elements.
- Robot kinematics refers to the geometry and movement of robotic mechanisms

Forward Kinematics (angles to position):

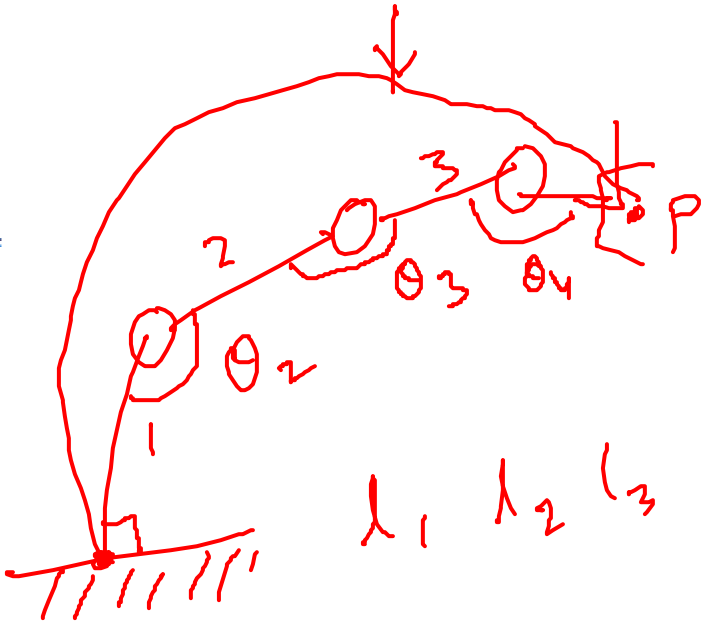
- What you are given: The length of each link, The angle of each joint
- What you can find: The position of any point (i.e. it’s (x, y, z) coordinates) end effector position and orientation



Given l_1, l_2, $t_{1,}$ t_2

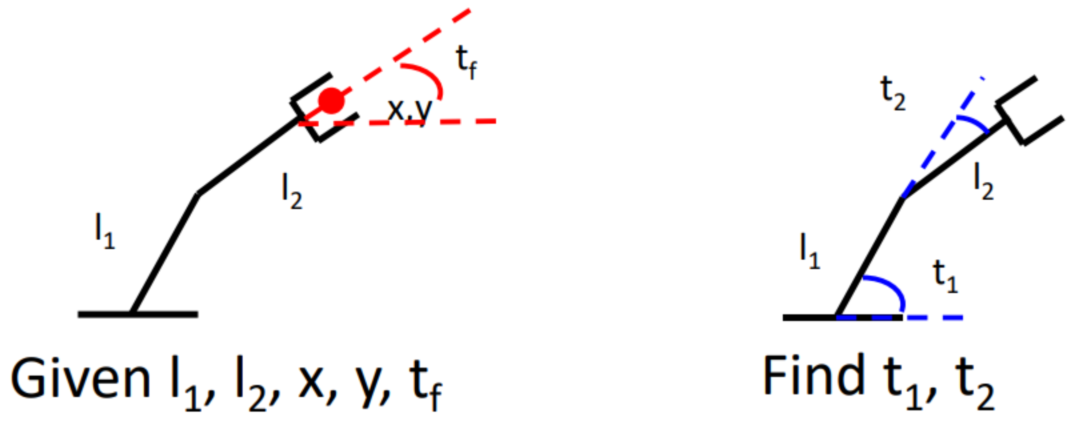


Find x, y, t_f



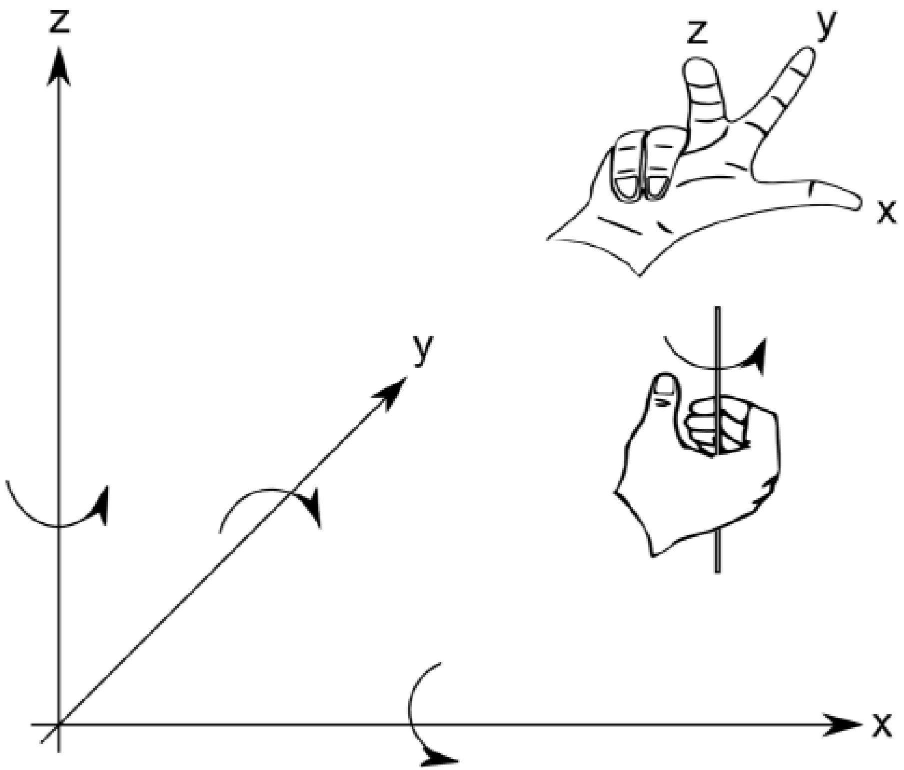
Inverse Kinematics (position to angles)

- What you are given: The length of each link, The position of some point on the robot
- What you can find: The angles of each joint needed to obtain that position



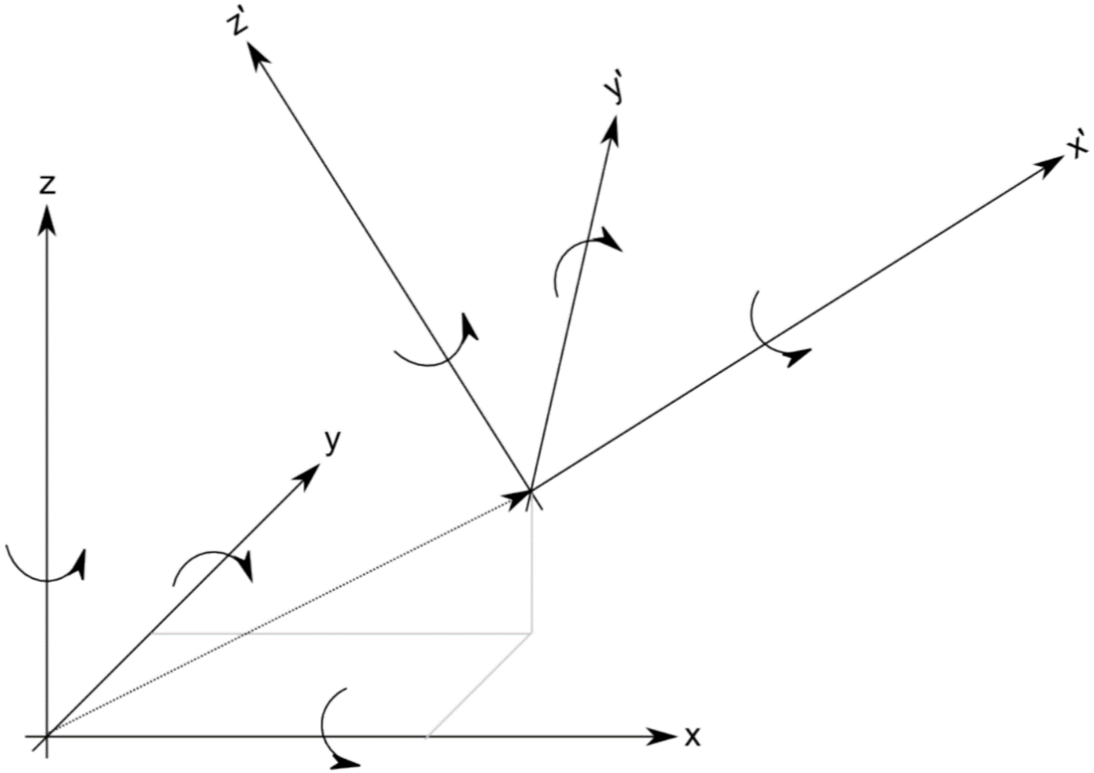
Coordinate Systems and Frames of Reference

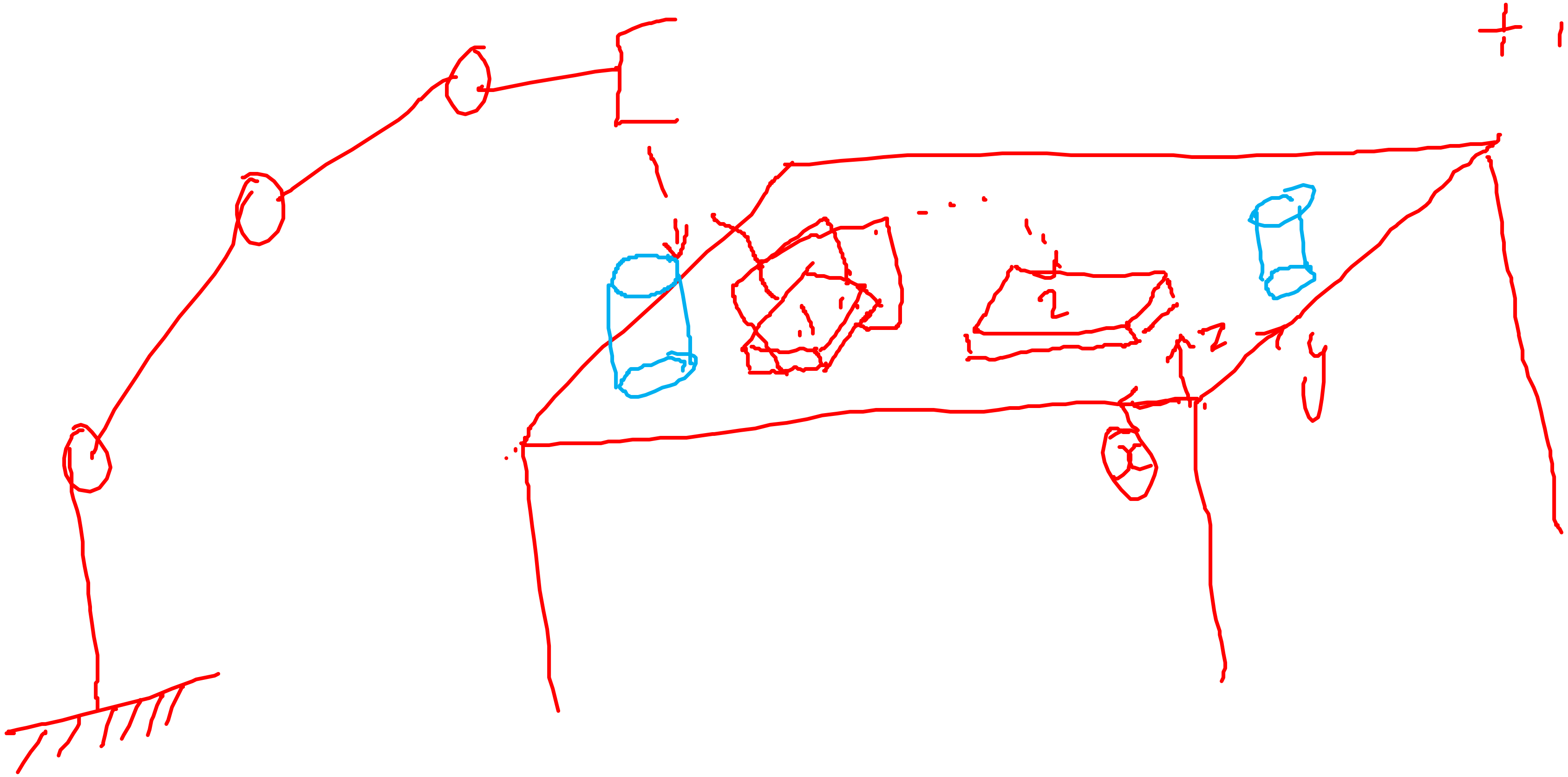
- Every robot assumes a position in the real world that can be described by its position (x, y and z) and orientation (pitch, yaw and roll) along the three major axes of a Cartesian Coordinate system



Coordinate Systems and Frames of Reference

- Defining all three position axes and orientations might be cumbersome.
- What level of detail we care about, where the origin of this coordinate system is, and even what kind of coordinate system we choose, depends on the specific application.
- For example, a simple mobile robot would typically require a representation with respect to a room, a building, or the earth's coordinate system (given by the longitude and latitude of each point on the earth), whereas a static manipulator usually has the origin of its coordinate system at its base.
- More complicated systems, such as mobile manipulators or multi-legged robots, make life much easier by defining multiple coordinate systems, e.g. one for each leg and one that describes the position of the robot in the world frame.
- These local coordinate systems are known as Frames of Reference.

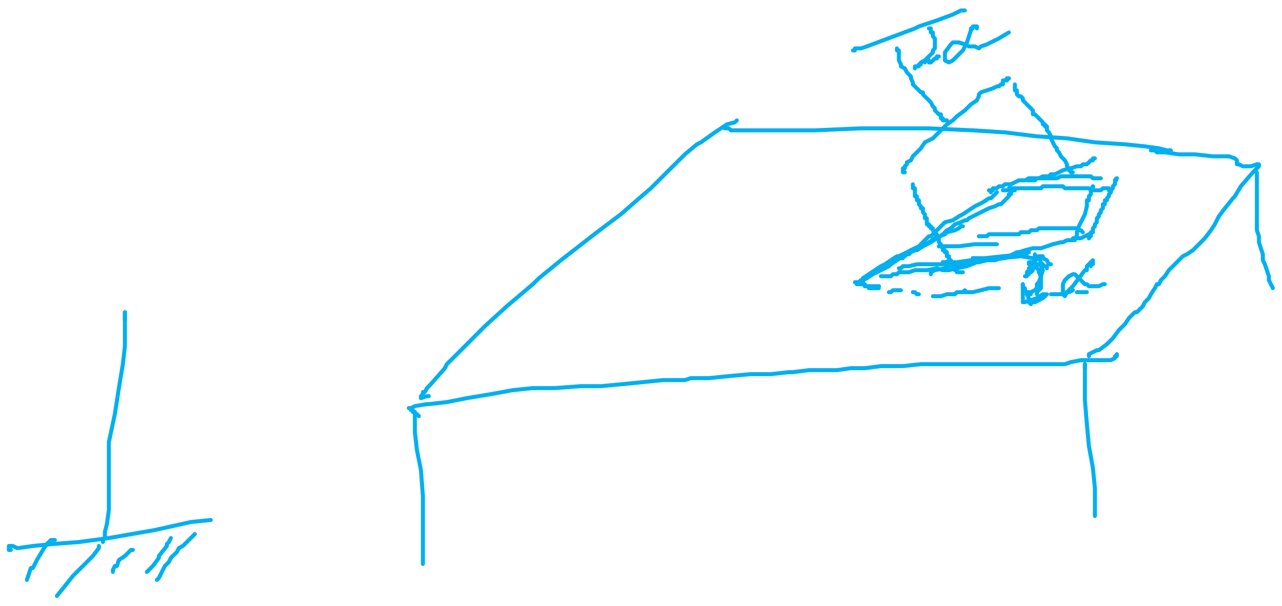




3 DOF

+ 1 DOF

5 DOF

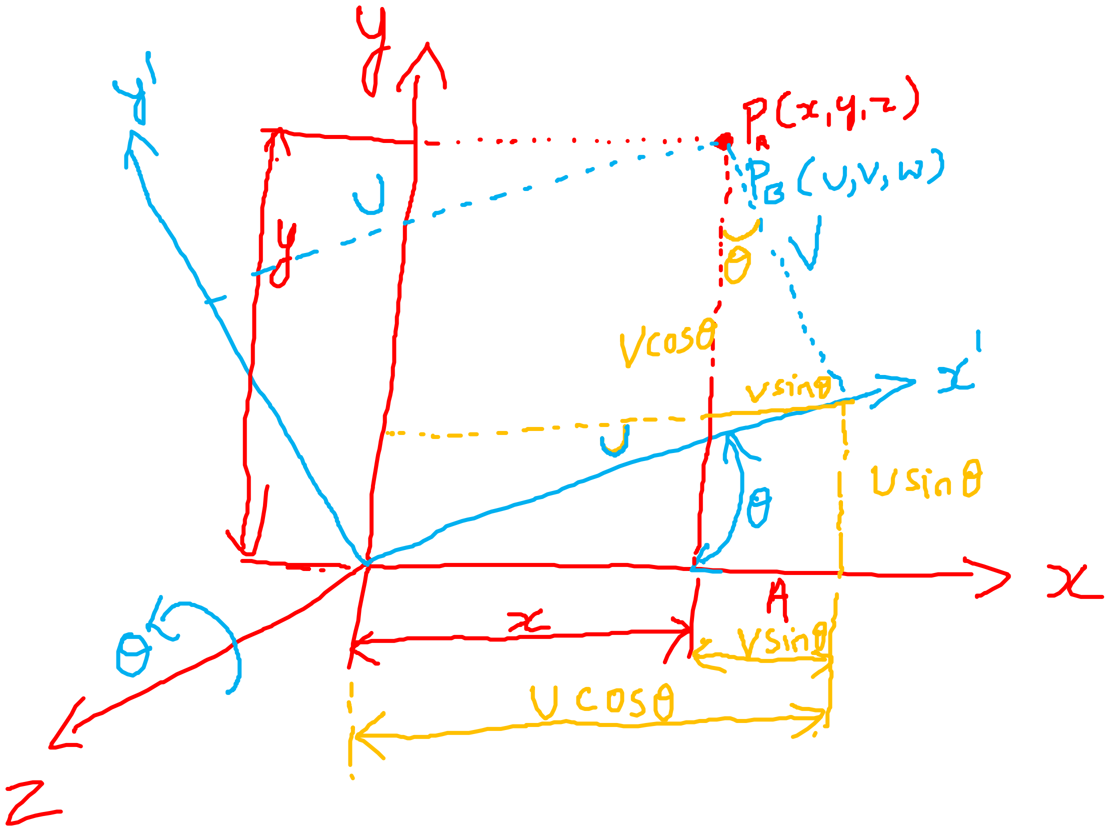


Transformations

- The relation between the body attached frame with the base frame of reference is described by the transformation matrix.
- The transformation is represented by following component transformations.

1. Rotation matrix
2. Translation matrix
3. Perspective transformation
4. Scaling

Rotational Matrix



$$x = u\cos\theta - v\sin\theta$$
$$y = u\sin\theta + v\cos\theta$$
$$z = w$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} u \\ v \\ w \end{Bmatrix}$$

Rotation Matrices in 3D

$$\mathbf{R}_z = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation around the Z-Axis

$R^{-1} = R^T$

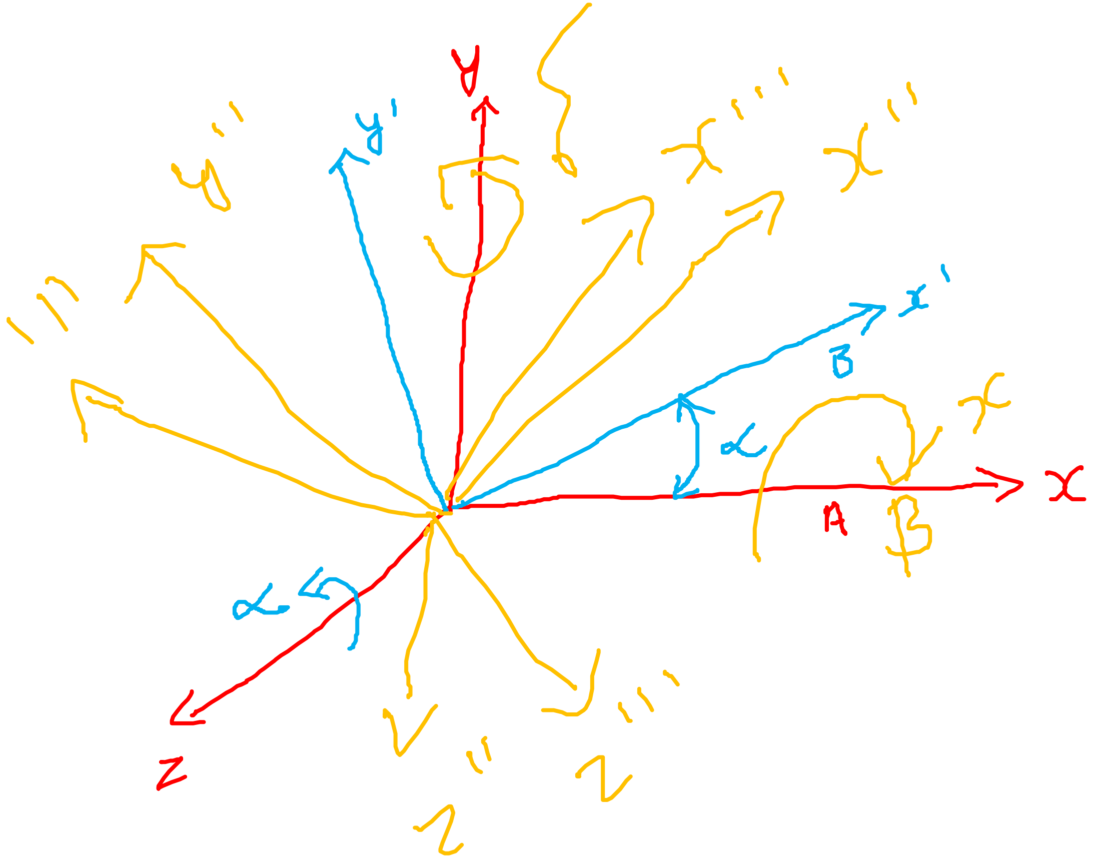
$$\mathbf{R}_y = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

Rotation around the Y-Axis

$$\mathbf{R}_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

Rotation around the X-Axis

Combined Rotational matrix



A & B are frames

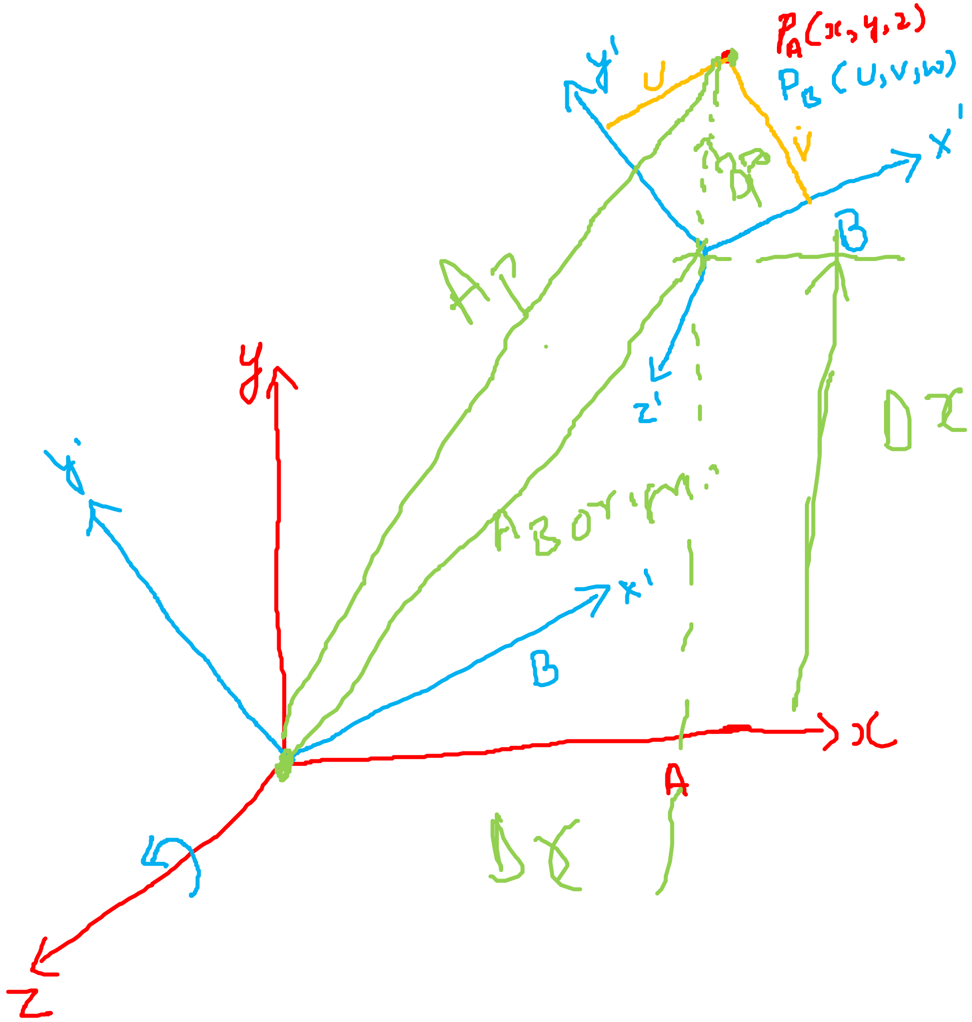
$\underbrace{B(\alpha, z)}^{1}$ $\underbrace{B(\beta, x)}^{2}$ $\underbrace{B(\gamma, y)}^{3}$

"Pre multiply"

$3 \times 2 \times 1$

$R[\gamma, y] \times R[\beta, x] \times R[\alpha, z]$

Rotational and Translation



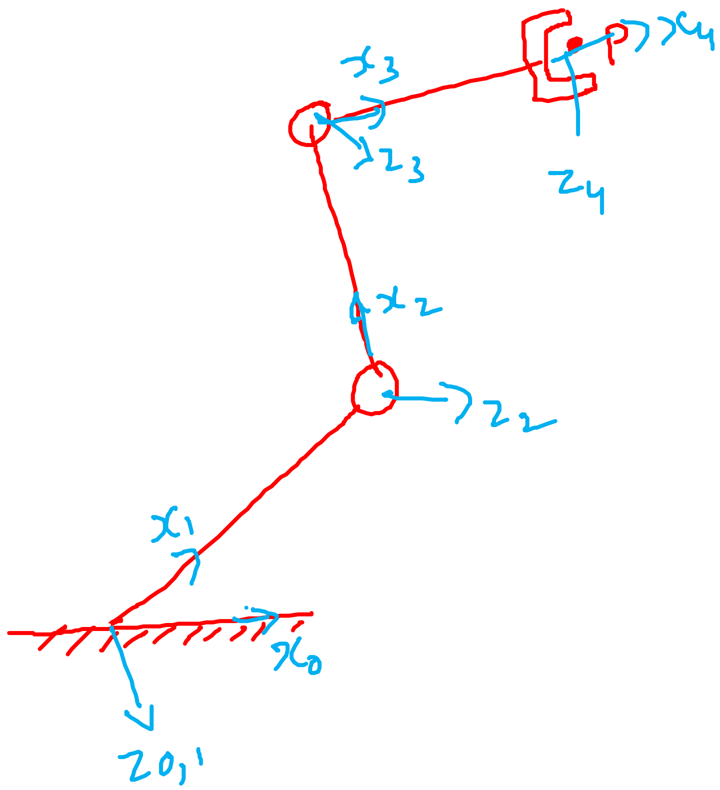
$$A_P = A_{B\,or\,m} + [R][B_P]$$

$$H = \begin{Bmatrix} x \\ y \\ z \\ 1 \end{Bmatrix} = \begin{bmatrix} c\theta & -s\theta & 0 & \Delta x \\ s\theta & c\theta & 0 & \Delta y \\ 0 & 0 & 1 & \Delta z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} U \\ V \\ W \\ 1 \end{Bmatrix}$$

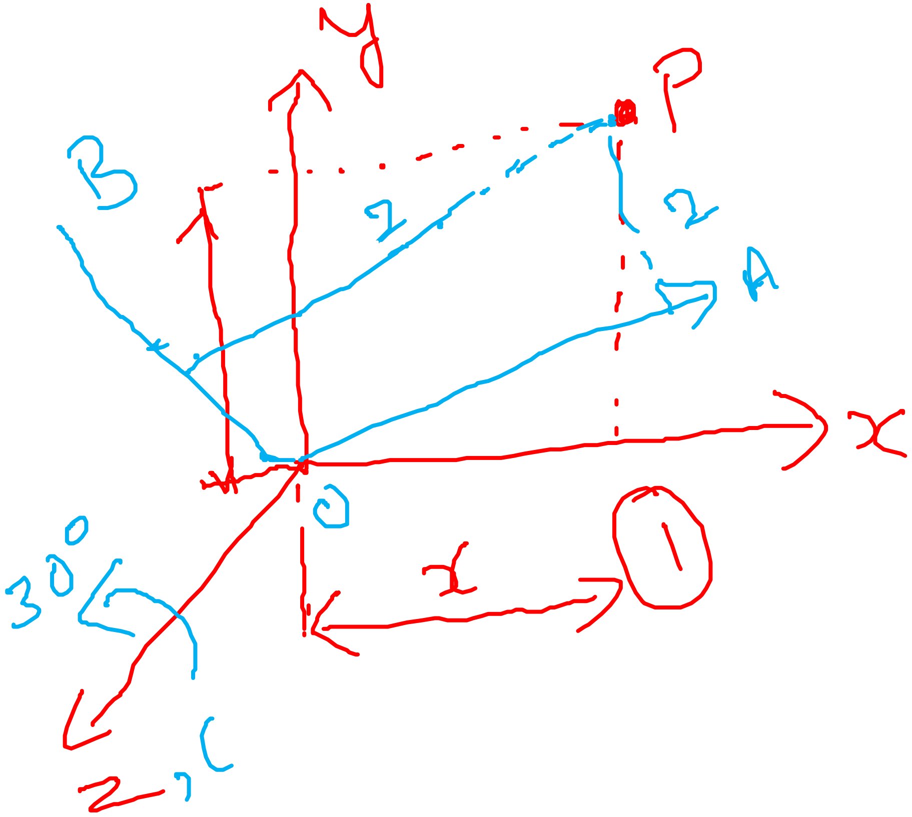
Homogeneous Transformation Matrix

$$
\underset{(4 \times 4)}{H} = \left[\begin{array}{c|c} \text{Rotation Matrix} & \text{Position Vector} \\ (3 \times 3) & (3 \times 1) \\ \hline \text{Perspective Transformation} & \text{scale factor} \\ (1 \times 3) & (1 \times 1) \end{array}\right]
$$

$H^{-1} \neq H^T$


x3
x4
z3
z4
x2
z2
x1
x0
z0,1

- The co-ordinate of a point $P_{abc}=(2, 2, 2)^T$ in the body co-ordinate frame OABC is rotated 30^0 about OZ axis. Determine the co-ordinates of the vector P_{xyz} with respect to base reference co-ordinate frame



$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} 2 \\ 2 \\ 2 \end{Bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{Bmatrix} 0.72 \\ 2.72 \\ 2 \end{Bmatrix}$$

- The co-ordinate of a point $P_{xyz}=(0.72, 2.72, 2)^T$ is given with respect to base reference co-ordinate frame. If body co-ordinate frame OABC is rotated 30^0 about OZ axis. Determine the co-ordinates of the vector P_{abc}

$$\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{bmatrix} C\theta & -S\theta & 0\\ S\theta & C\theta & 0\\ 0 & 0 & 1\end{bmatrix}\begin{Bmatrix}u\\v\\w\end{Bmatrix}$$

↓ given ↓ find

$$(x) = [R](u)$$

$$R^{-1} = R^T$$

$$u = (R^{-1})(x)$$

$$\begin{Bmatrix}u\\v\\w\end{Bmatrix} = \begin{bmatrix} C\theta & S\theta & 0\\ -S\theta & C\theta & 0\\ 0 & 0 & 1\end{bmatrix}\begin{Bmatrix}x\\y\\z\end{Bmatrix} = \begin{Bmatrix}2\\2\\2\end{Bmatrix}$$

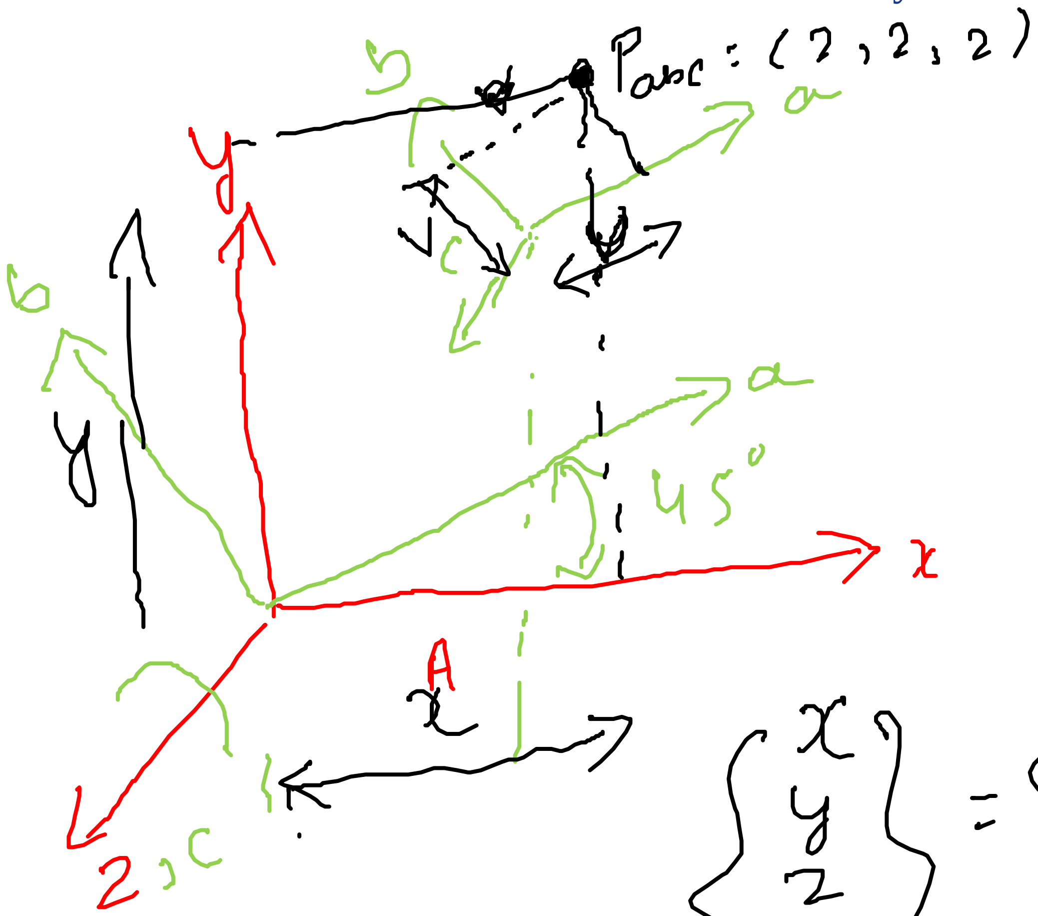
- Frame B is rotated by 30^0 about X, then by 60^0 about Y and then 30^0 about Z. P_B=(1,1,1) find P_A

$$\begin{Bmatrix} x \\ y \\ z \end{Bmatrix} = \underbrace{\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}}_{} \times \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \times \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \begin{Bmatrix} 1 \\ 1 \\ 1 \end{Bmatrix}$$

R_z, R_y, R_x

$$\begin{Bmatrix} x \\ y \\ z \end{Bmatrix} = \begin{Bmatrix} 1.25 \\ 1.14 \\ -0.18 \end{Bmatrix}$$

- The co-ordinate of a point $P_{abc}=(2, 2, 2)^T$ in the body co-ordinate frame OABC is rotated by 45^0 about Z axis of frame A and Translated by(5,5,5) units in xyz direction of A. Determine the co-ordinates of the vector P_{xyz} with respect to base reference co-ordinate frame



$P(x, y z)$

$$\begin{Bmatrix} x \\ y \\ z \\ 1 \end{Bmatrix} = \begin{bmatrix} C\theta & -S\theta & 0 & 5 \\ S\theta & C\theta & 0 & 5 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} 2 \\ 2 \\ 2 \\ 1 \end{Bmatrix}$$

$$\begin{Bmatrix} x \\ y \\ z \\ 1 \end{Bmatrix} = \begin{Bmatrix} 5 \\ 7.8 \\ 7 \\ 1 \end{Bmatrix}$$

$x = 5;\ y = 7.8;\ z = 7$

- If P_A =(5, 7.8, 7), Translated by (5, 5, 5,), rotated by 45^0 about Z find P_B

$$\{U\} = (x)\,[R_T]^{-1}$$

$$\begin{Bmatrix} x \\ y \\ z \\ 1 \end{Bmatrix} = \underbrace{\begin{bmatrix} c\theta & -s\theta & 0 & \Delta x \\ s\theta & c\theta & 0 & \Delta y \\ 0 & 0 & 1 & \Delta z \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{R_T} \begin{Bmatrix} u \\ v \\ w \\ 1 \end{Bmatrix} \rightarrow \text{find}$$

$$[R]^{-1} = R^T$$

$$\Rightarrow T^{-1} = -R^T \begin{Bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{Bmatrix}$$

$$\begin{Bmatrix} u \\ v \\ w \\ 1 \end{Bmatrix} = \begin{bmatrix} c\theta & s\theta & 0 & -7 \\ -s\theta & c\theta & 0 & 0 \\ 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} 5 \\ 7.8 \\ 7 \\ 1 \end{Bmatrix} = \begin{Bmatrix} 2 \\ 2 \\ 2 \\ 1 \end{Bmatrix}$$

$$= \begin{bmatrix} c\theta & s\theta & 0 \\ -s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} -5 \\ -5 \\ -5 \end{Bmatrix}$$

$$= \begin{bmatrix} -7 \\ 0 \\ -5 \end{bmatrix}$$

- A point $P_{abc}=(2,3,4)^T$ has to be translated through distance of +4units along OX axis and -2 units along OZ axis. Determine the co-ordinates of the new point P_{xyz} by homogeneous transformation.

$$\begin{Bmatrix} x \\ y \\ z \\ 1 \end{Bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{Bmatrix} =$$